1.42 Trigonometry: Trigonometric Identities in Degrees

Worked Solutions

1 a)

$\sin(x + 180°) = -\sin x.$

1 b)

$\sin(-x) = -\sin x.$

1 c)

$\sin(x + 360°) = \sin x.$

1 d)

$\sin(x - 180°) = -\sin x.$

1 e)

$\sin(x - 360°) = \sin x.$

1 f)

$\sin(180° - x) = \sin x.$

1 g)

$\sin(360° - x) = -\sin x.$

1 h)

$\sin(540° - x) = \sin x.$

1 i)

$\sin(x - 540°) = -\sin x.$

1 j)

$\sin(x + 540°) - \sin x.$

1 k)

$\sin(x + 720°) = \sin x.$

1 l)

$\sin(900° - x) = \sin x.$

1 m)

$\sin(3600° - x) = -\sin x.$

2 a)

$\cos(x - 180°) = -\cos x.$

2 b)

$\cos(-x) = \cos x.$

2 c)

$\cos(180° - x) = -\cos x.$

2 d)

$\cos(x + 180°) = -\cos x.$

2 e)

$\cos(x + 360°) = \cos x.$

2 f)

$\cos(360° - x) = \cos x.$

2 g)

$\cos(x - 360°) = \cos x.$

2 h)

$\cos(540° - x) = -\cos x.$

2 i)

$\cos(x + 540°) = -\cos x.$

2 j)

$\cos(x - 540°) = -\cos x.$

2 k)

$\cos(x - 720°) = \cos x.$

2 l)

$\cos(900° - x) = -\cos x.$

2 m)

$\cos(360° - x) = \cos x.$

3 a)

$\tan(x + 180°) = \tan x.$

3 b)

$\tan(-x) = -\tan x.$

3 c)

$\tan(x + 360°) = \tan x.$

3 d)

$\tan(x - 180°) = \tan x.$

3 e)

$\tan(360° - x) = -\tan x.$

3 f)

$\tan(180° - x) = -\tan x.$

3 g)

$\tan(x - 360°) = \tan x.$

3 h)

$\tan(540° - x) = -\tan x.$

3 i)

$\tan(x - 540°) = \tan x.$

3 j)

$\tan(x + 540°) = \tan x.$

3 k)

$\tan(x + 720°) = \tan x.$

3 l)

$\tan(900° - x) = -\tan x.$

3 m)

$\tan(3600° - x) = -\tan x.$

4 a)

$\sin(x + 90°) = \cos x.$

4 b)

$\sin(x - 90°) = -\cos x.$

4 c)

$\sin(x + 270°) = -\cos x.$

4 d)

$\sin(90° - x) = \cos x.$

4 e)

$\sin(270° - x) = -\cos x.$

4 f)

$\sin(x + 450°) = \cos x.$

4 g)

$\sin(x - 450°) = -\cos x.$

4 h)

$\sin(450° - x) = \cos x.$

4 i)

$\sin(x - 630°) = \cos x.$

4 j)

$\sin(630° - x) = -\cos x.$

4 k)

$\sin(x + 810°) = \cos x.$

4 l)

$\sin(990° - x) = -\cos x.$

4 m)

$\sin(9090° - x) = \cos x.$

5 a)

$\cos(x + 90°) = -\sin x.$

5 b)

$\cos(x - 90°) = \sin x.$

5 c)

$\cos(x + 270°) = \sin x.$

5 d)

$\cos(90^\circ - x) = \sin x.$

5 e)

$\cos(270^\circ - x) = -\sin x.$

5 f)

$\cos(x + 450^\circ) = -\sin x.$

5 g)

$\cos(x - 450^\circ) = \sin x.$

5 h)

$\cos(450^\circ - x) = \sin x.$

5 i)

$\cos(x - 630^\circ) = -\sin x.$

5 j)

$\cos(630^\circ - x) = -\sin x.$

5 k)

$\cos(x + 810^\circ) = -\sin x.$

5 l)

$\cos(990^\circ - x) = -\sin x.$

5 m)

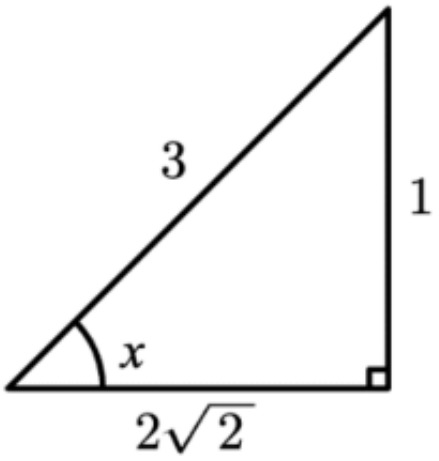
$\cos(9450^\circ - x) = \sin x.$

6 a)

$\sin x = \frac{1}{3}$.

x is **ACUTE**.

Draw a triangle with this information:



$\cos x = \frac{2\sqrt{2}}{3}$.

6 b)

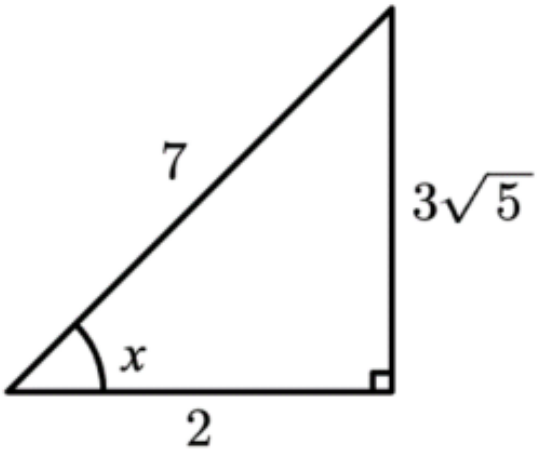
$\tan x = \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4}$.

7 a)

$\cos x = \frac{2}{7}$.

x is **ACUTE**.

Draw a triangle with this information:



$\sin x = \frac{3\sqrt{5}}{7}$.

7 b)

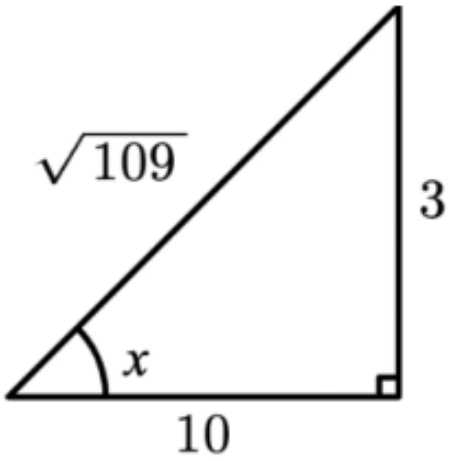
$\tan x = \frac{3\sqrt{5}}{2}$.

8 a)

$\tan x = \frac{3}{10}$.

x is **ACUTE**.

Draw a triangle with this information:



$\sin x = \frac{3}{\sqrt{109}} = \frac{3\sqrt{109}}{109}$.

8 b)

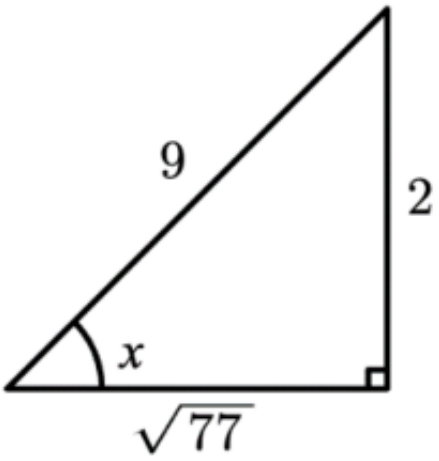
$\cos x = \frac{10}{\sqrt{109}} = \frac{10\sqrt{109}}{109}$.

9 a)

$\sin x = \frac{2}{9}$.

x is **OBTUSE**.

Ignore $90° < x < 180°$ for the moment and draw a right-angled triangle with this information:



$\cos x = -\frac{\sqrt{77}}{9}$ (the result is negative as $\cos x$ is below the x-axis between 90° and 180°).

9 b)

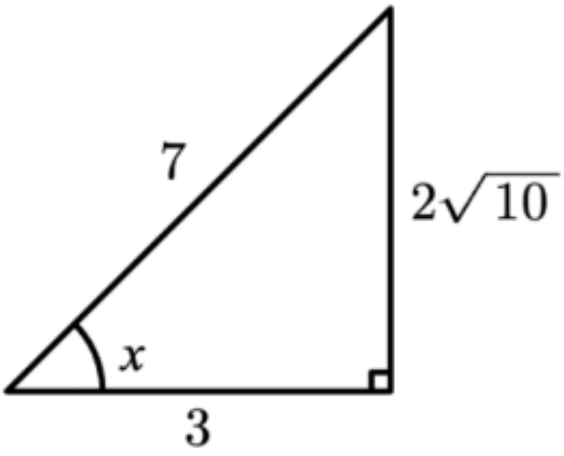
$\tan x = -\frac{2}{\sqrt{77}} = -\frac{2\sqrt{77}}{77}$ (the result is negative as $\tan x$ is below the x-axis between 90° and 180°).

10 a)

$\cos x = -\frac{3}{7}$.

x is **OBTUSE**.

Ignore $90° < x < 180°$ and the negative sign for the moment and draw a right-angled triangle with this information:



$\sin x = \frac{2\sqrt{10}}{7}$ (the result is positive as $\sin x$ is above the x-axis between 90° and 180°).

10 b)

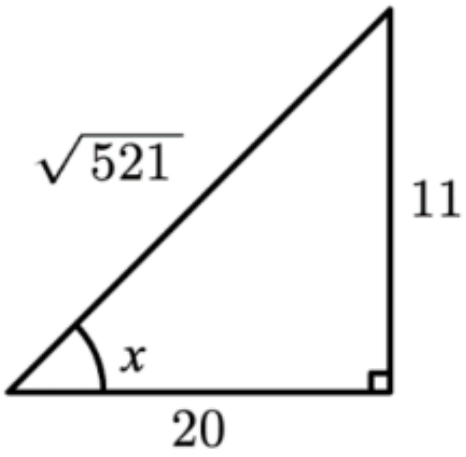
$\tan x = -\frac{2\sqrt{10}}{3}$ (the result is negative as $\tan x$ is below the x-axis between $90°$ and $180°$).

11 a)

$\tan x = -\frac{11}{20}$.

x is **OBTUSE**.

Ignore $90° < x < 180°$ and the negative sign for the moment and draw a right-angled triangle with this information:



$\sin x = \frac{11}{\sqrt{521}} = \frac{11\sqrt{521}}{521}$ (the result is positive as $\sin x$ is above the x-axis between $90°$ and $180°$).

11 b)

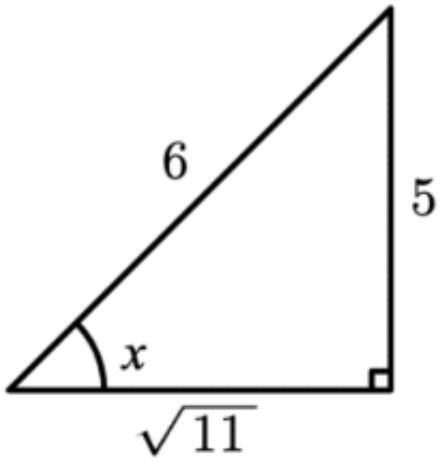
$\cos x = -\frac{20}{\sqrt{521}} = -\frac{20\sqrt{521}}{521}$ (the result is negative as $\cos x$ is below the x-axis between $90°$ and $180°$).

12 a)

$\sin x = -\frac{5}{6}$.

$180° < x < 270°$.

Ignore $180° < x < 270°$ and the negative sign for the moment and draw a right-angled triangle with this information:



$\cos x = -\frac{\sqrt{11}}{6}$ (the result is negative as $\cos x$ is below the x-axis between 180° and 270°).

12 b)

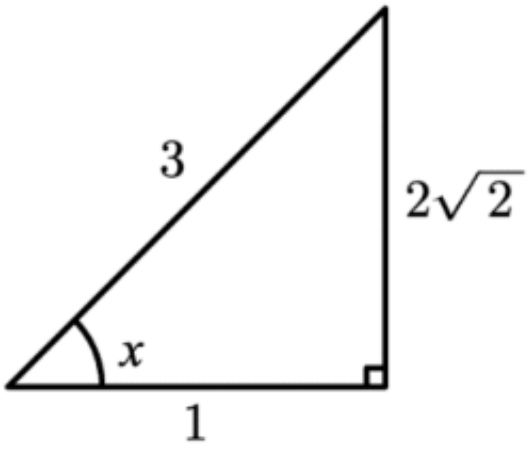
$\tan x = \frac{5}{\sqrt{11}} = \frac{5\sqrt{11}}{11}$ (the result is positive as $\tan x$ is above the x-axis between 180° and 270°).

13 a)

$\cos x = -\frac{1}{3}$.

$180° < x < 270°$.

Ignore $180° < x < 270°$ and the negative sign for the moment and draw a right-angled triangle with this information:



$\sin x = -\frac{2\sqrt{2}}{3}$ (the result is negative as $\sin x$ is below the x-axis between 180° and 270°).

13 b)

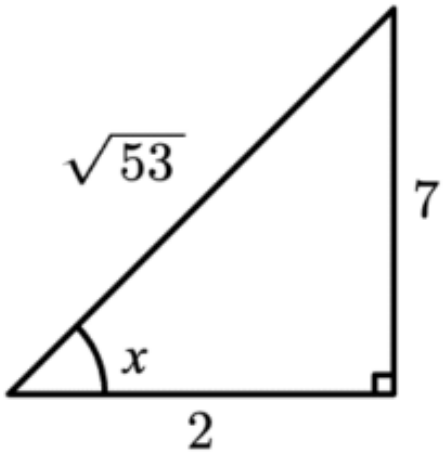
$\tan x = \frac{2\sqrt{2}}{1} = 2\sqrt{2}$ (the result is positive as $\tan x$ is above the x-axis between 180° and 270°).

14 a)

$\tan x = \frac{7}{2}$.

$180° < x < 270°$.

Ignore $180° < x < 270°$ for the moment and draw a right-angled triangle with this information:



$\sin x = -\frac{7}{\sqrt{53}} = -\frac{7\sqrt{53}}{53}$ (the result is negative as $\sin x$ is below the x-axis between 180° and 270°).

14 b)

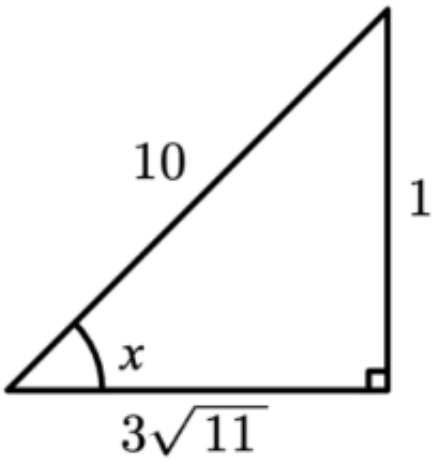
$\cos x = -\frac{2}{\sqrt{53}} = -\frac{2\sqrt{53}}{53}$ (the result is negative as $\cos x$ is below the x-axis between 180° and 270°).

15 a)

$\sin x = -\frac{1}{10}$.

$270° < x < 360°$.

Ignore $270° < x < 360°$ and the negative sign for the moment and draw a right-angled triangle with this information:



$\cos x = \frac{3\sqrt{11}}{10}$ (the result is positive as $\cos x$ is above the x-axis between 270° and 360°).

15 b)

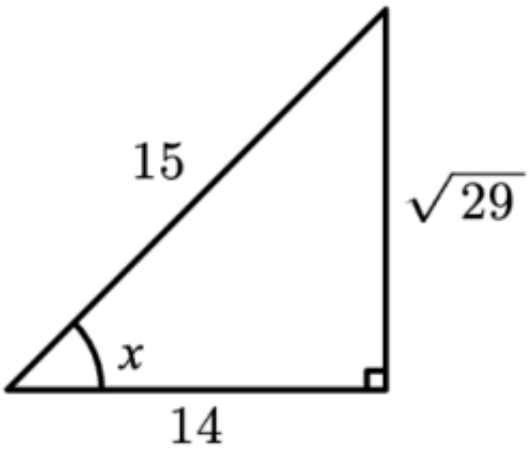
$\tan x = -\frac{1}{3\sqrt{11}} = -\frac{\sqrt{11}}{33}$ (the result is negative as $\tan x$ is below the x-axis between 270° and 360°).

16 a)

$\cos x = \frac{14}{15}$.

$270° < x < 360°$.

Ignore $270° < x < 360°$ for the moment and draw a right-angled triangle with this information:



$\sin x = -\frac{\sqrt{29}}{15}$ (the result is negative as $\sin x$ is below the x-axis between 270° and 360°).

16 b)

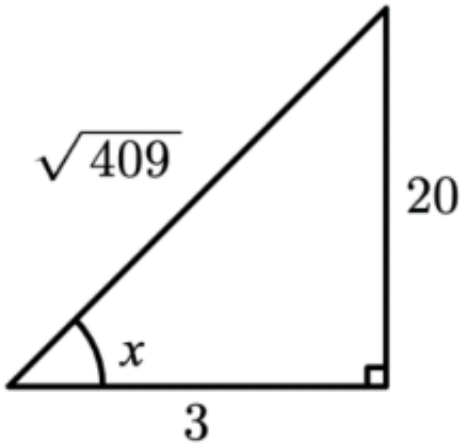
$\tan x = -\frac{\sqrt{29}}{14}$ (the result is negative as $\tan x$ is below the x-axis between 270° and 360°).

17 a)

$\tan x = -\frac{20}{3}$.

$270° < x < 360°$.

Ignore $270° < x < 360°$ and the negative sign for the moment and draw a right-angled triangle with this information:



$\sin x = -\frac{20}{\sqrt{409}} = -\frac{20\sqrt{409}}{409}$ (the result is negative as $\sin x$ is below the x-axis between 270° and 360°).

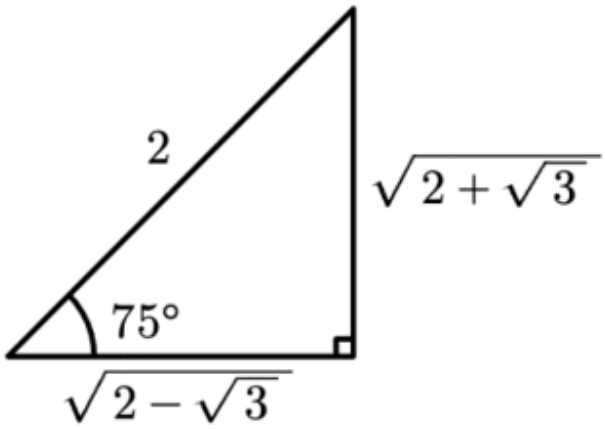
17 b)

$\cos x = \frac{3}{\sqrt{409}} = \frac{3\sqrt{409}}{409}$ (the result is positive as $\cos x$ is above the x-axis between 270° and 360°).

18 a)

$\cos 75° = \frac{\sqrt{2-\sqrt{3}}}{2} = \frac{\text{Adjacent}}{\text{Hypotenuse}}$.

$\text{Opposite} = \sqrt{2^2 - (\sqrt{2-\sqrt{3}})^2} = \sqrt{4-(2-\sqrt{3})} = \sqrt{2+\sqrt{3}}$.



$\sin 75° = \frac{\sqrt{2+\sqrt{3}}}{2}$.

18 b)

$$\tan 75° = \frac{\sqrt{2+\sqrt{3}}}{\sqrt{2-\sqrt{3}}} = \sqrt{\frac{2+\sqrt{3}}{2-\sqrt{3}}} = \sqrt{\frac{2+\sqrt{3}}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}}}$$
$$= \sqrt{\frac{4+4\sqrt{3}+3}{4-3}} = \sqrt{7+4\sqrt{3}}$$

19 a)

$\cos x \tan x \equiv \cos x \times \frac{\sin x}{\cos x} \equiv \sin x$.

19 b)

$\sin x \tan x \equiv \sin x \times \frac{\sin x}{\cos x} \equiv \frac{\sin^2 x}{\cos x}$.

19 c)

$\frac{\tan x}{\sin x} \equiv \frac{\frac{\sin x}{\cos x}}{\sin x} \equiv \frac{\sin x}{\cos x} \times \frac{1}{\sin x} = \frac{1}{\cos x}$.

19 d)

$\frac{\tan x}{\cos x} \equiv \frac{\frac{\sin x}{\cos x}}{\cos x} \equiv \frac{\sin x}{\cos x} \times \frac{1}{\cos x} \equiv \frac{\sin x}{\cos^2 x}$.

20 a)

$\sin x \cos x \tan x \equiv \sin x \cos x \times \frac{\sin x}{\cos x} \equiv \sin^2 x$.

20 b)

$\sin x \cos^2 x \tan^2 x \equiv \sin x \cos^2 x \times \frac{\sin^2 x}{\cos^2 x} \equiv \sin^3 x.$

20 c)

$$\begin{aligned}\frac{\sin x}{\cos x \tan^2 x} &\equiv \frac{\sin x}{\cos x \times \frac{\sin^2 x}{\cos^2 x}} \equiv \frac{\sin x}{\frac{\sin^2 x}{\cos x}} \equiv \sin x \times \frac{\cos x}{\sin^2 x} \\ &\equiv \frac{\cos x}{\sin x}\end{aligned}$$

Alternatively:

$\frac{\sin x}{\cos x \tan^2 x} \equiv \frac{\frac{\sin x}{\cos x}}{\tan^2 x} \equiv \frac{\tan x}{\tan^2 x} \equiv \frac{1}{\tan x} \equiv \frac{\cos x}{\sin x}.$

20 d)

$\frac{\tan^2 x}{\sin x \cos x} \equiv \frac{\frac{\sin^2 x}{\cos^2 x}}{\sin x \cos x} \equiv \frac{\sin^2 x}{\cos^2 x} \times \frac{1}{\sin x \cos x} \equiv \frac{\sin x}{\cos^3 x}.$

21 a)

$1 - \cos^2 x \equiv \sin^2 x.$

21 b)

$\sqrt{1 - \sin^2 x} \equiv \sqrt{\cos^2 x} \equiv |\cos x|.$

21 c)

$9 \cos^2 x - 9 \equiv 9(\cos^2 x - 1) \equiv -9(1 - \cos^2 x) = -9 \sin^2 x.$

21 d)

$8 \cos^2 x + 8 \sin^2 x \equiv 8(\cos^2 x + \sin^2 x) = 8 \times 1 \equiv 8.$

22 a)

$\frac{1-\sin^2 x}{\cos^2 x} \equiv \frac{\cos^2 x}{\cos^2 x} \equiv 1.$

22 b)

$\frac{1-\cos^2 x}{\sin^2 x} \equiv \frac{\sin^2 x}{\sin^2 x} \equiv 1.$

22 c)

$\frac{4-4\cos^2 x}{5-5\sin^2 x} \equiv \frac{4(1-\cos^2 x)}{5(1-\sin^2 x)} \equiv \frac{4 \sin^2 x}{5 \cos^2 x} \equiv \frac{4}{5} \tan^2 x.$

22 d)

$$\begin{aligned}\frac{\tan x}{6 - \frac{6}{\cos^2 x}} &\equiv \frac{\tan x \cos^2 x}{6 \cos^2 x - 6} \equiv \frac{\frac{\sin x}{\cos x} \times \cos^2 x}{6(\cos^2 x - 1)} \equiv -\frac{\frac{\sin x}{\cos x} \times \cos^2 x}{6(1 - \cos^2 x)} \\ &\equiv -\frac{\sin x \cos x}{6 \sin^2 x} \equiv -\frac{\cos x}{6 \sin x} \equiv -\frac{1}{6} \times \frac{\cos x}{\sin x} \equiv -\frac{1}{6} \times \frac{1}{\frac{\sin x}{\cos x}} \\ &\equiv -\frac{1}{6 \tan x}\end{aligned}$$

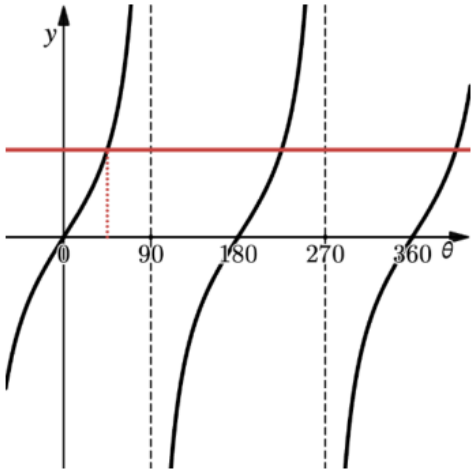
23 a)

$\sin\theta = \cos\theta$ for $0° \leq \theta < 360°$.

$\Rightarrow \frac{\sin\theta}{\cos\theta} = 1 \Rightarrow \tan\theta = 1$.

Find first value:

$\theta = \arctan 1 = 45°$ (in range).



Add period:

$\theta = 45° + 180° = 225°$ (in range).

$\Rightarrow \theta = \{45°, 225°\}$.

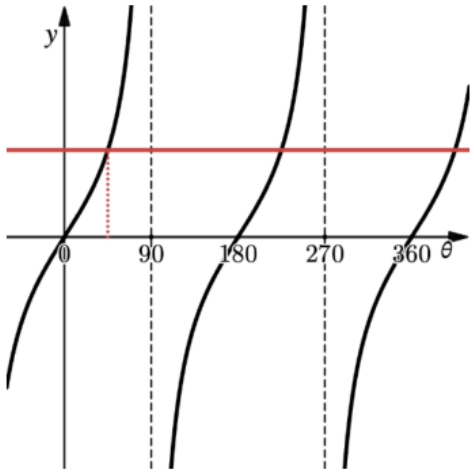
23 b)

$2\sin\theta = 3\cos\theta$ for $0° \leq \theta < 360°$.

$\Rightarrow \frac{\sin\theta}{\cos\theta} = \frac{3}{2} \Rightarrow \tan\theta = \frac{3}{2}$.

Find first value:

$\theta = \arctan\frac{3}{2} = 56.309\ldots°$ (in range).



Add period:

$\theta = 56.309\ldots° + 180° = 236.309\ldots°$ (in range).

$\Rightarrow \theta = \{56.3°, 236°\}$ (3 SF).

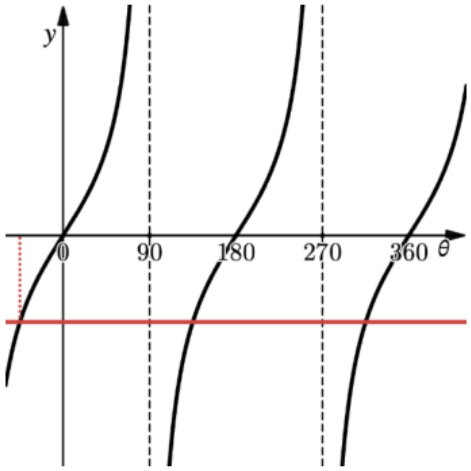
23 c)

$8\sin\theta + 5\cos\theta = 0$ for $0° \leq \theta < 360°$.

$\Rightarrow 8\sin\theta = -5\cos\theta \Rightarrow \frac{\sin\theta}{\cos\theta} = -\frac{5}{8} \Rightarrow \tan\theta = -\frac{5}{8}$.

Find first value:

$\theta = \arctan(-\frac{5}{8}) = -32.005\ldots°$ (not in range).



Add on period:

$\theta = -32.005\ldots° + 180° = 147.994\ldots°$ (in range).

$\theta = 147.994\ldots° + 180° = 327.994\ldots°$ (in range).

$\Rightarrow \theta = \{148°, 328°\}$ (3 SF).

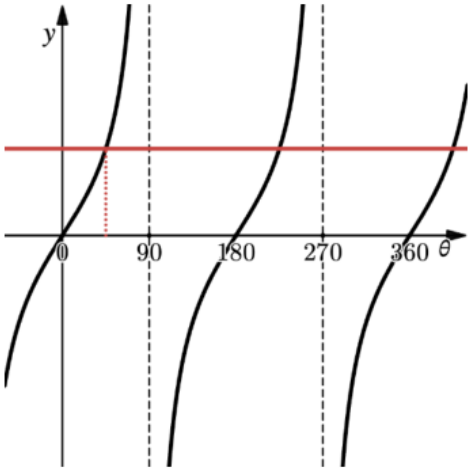
23 d)

$7\cos\theta - 2\sin\theta = 0$ for $0° \leq \theta < 360°$.

$\Rightarrow 2\sin\theta = 7\cos\theta \Rightarrow \frac{\sin\theta}{\cos\theta} = \frac{7}{2} \Rightarrow \tan\theta = \frac{7}{2}$.

Find first value:

$\theta = \arctan\frac{7}{2} = 74.054\ldots°$ (in range).

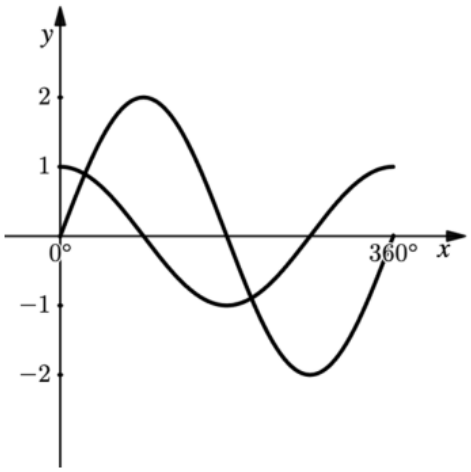


Add period:

$\theta = 74.054\ldots° + 180° = 254.054\ldots°$ (in range).

$\Rightarrow \theta = \{74.1°, 254°\}$ (3 SF).

24 a)



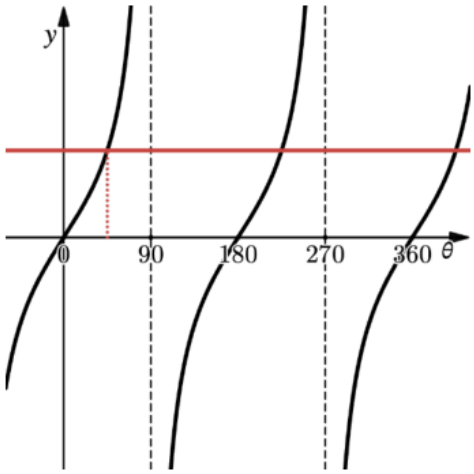
24 b)

$2\sin x = \cos x$ for $0° \leq x < 360°$.

$\Rightarrow \frac{\sin x}{\cos x} = \frac{1}{2} \Rightarrow \tan x = \frac{1}{2}$.

Find first value:

$x = \arctan \frac{1}{2} = 26.565\ldots°$ (in range).



Add period:

$x = 26.565\ldots° + 180° = 206.565\ldots°$ (in range).

$\Rightarrow x = \{26.6°, 207°\}$ (3 SF).

$(26.6°, 0.894)$ and $(207°, -0.894)$ (3 SF).

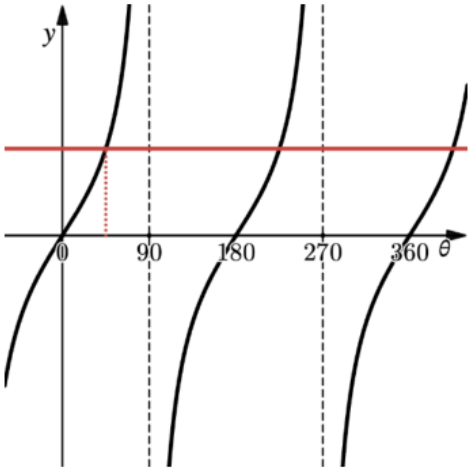
25 a)

$6\sin 3\theta = 5\cos 3\theta$ for $0° \leq \theta < 240°$.

$\Rightarrow \frac{\sin 3\theta}{\cos 3\theta} = \frac{5}{6} \Rightarrow \tan 3\theta = \frac{5}{6}$.

Find first value:

$3\theta = \arctan \frac{5}{6} = 39.805\ldots°$.



Apply transformation:

$3\theta = 39.805\ldots° \Rightarrow \theta = 13.268\ldots°$ (in range).

Add period:

$\theta = 13.268\ldots° + 60° = 73.268\ldots°$ (in range).

$\theta = 73.268\ldots° + 60° = 133.268\ldots°$ (in range).

$\theta = 133.268\ldots° + 60° = 193.268\ldots°$ (in range).

$\theta = \{13.3°, 73.3°, 133°, 193°\}$ (3 SF).

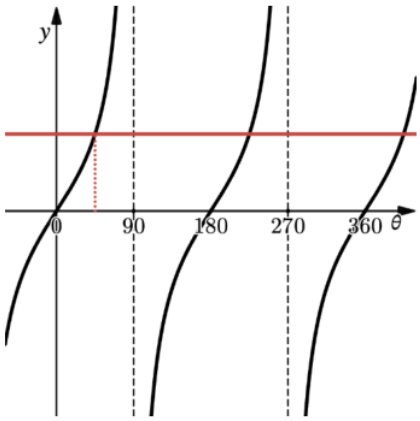
25 b)

$7 \sin \frac{1}{2}\theta = 9 \cos \frac{1}{2}\theta$ for $0° \leq \theta < 720°$.

$\Rightarrow \frac{\sin \frac{1}{2}\theta}{\cos \frac{1}{2}\theta} = \frac{9}{7} \Rightarrow \tan \frac{1}{2}\theta = \frac{9}{7}$.

Find first value:

$\frac{1}{2}\theta = \arctan \frac{9}{7} = 52.125\ldots°$.



Apply transformation:

$\frac{1}{2}\theta = 52.125\ldots° \Rightarrow \theta = 104.250\ldots°$ (in range).

Add period:

$\theta = 104.250\ldots° + 360° = 464.250\ldots°$ (in range).

$\theta = \{104°, 464°\}$ (3 SF).

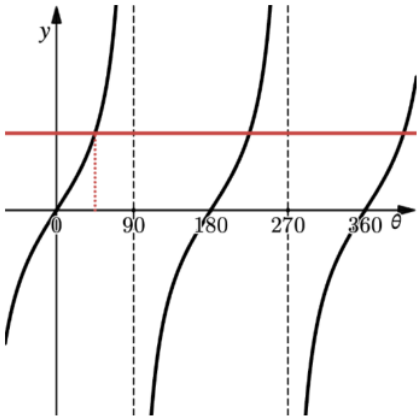
25 c)

$11\cos\frac{1}{3}\theta - 25\sin\frac{1}{3}\theta = 0$ for $0° \leq \theta < 2160°$.

$\Rightarrow 25\sin\frac{1}{3}\theta = 11\cos\frac{1}{3}\theta \Rightarrow \frac{\sin\frac{1}{3}\theta}{\cos\frac{1}{3}\theta} = \frac{11}{25} \Rightarrow \tan\frac{1}{3}\theta = \frac{11}{25}$.

Find first value:

$\frac{1}{3}\theta = \arctan\frac{11}{25} = 23.749\ldots°$.



Apply transformation:

$\frac{1}{3}\theta = 23.749\ldots° \Rightarrow \theta = 71.248\ldots°$ (in range).

Add period:

$\theta = 71.248\ldots° + 540° = 611.248\ldots°$ (in range).

$\theta = 611.248\ldots° + 540° = 1151.248\ldots°$ (in range).

$\theta = 1151.248\ldots° + 540° = 1691.248\ldots°$ (in range).

$\theta = \{71.2°, 611°, 1150°, 1690°\}$ (3 SF).

26 a)

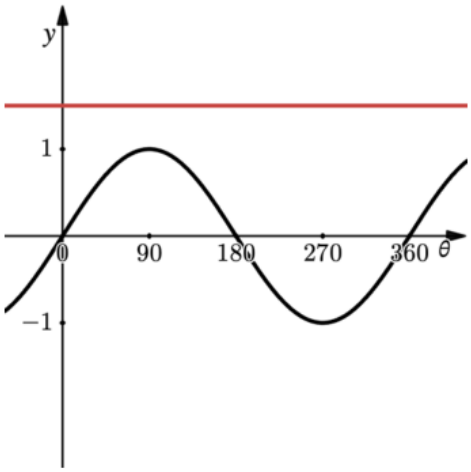
$4\tan\theta + \cos\theta = 0$ for $0° \leq \theta < 360°$.

$\Rightarrow \frac{4\sin\theta}{\cos\theta} + \cos\theta = 0 \Rightarrow 4\sin\theta + \cos^2\theta = 0.$

$\Rightarrow 4\sin\theta + (1 - \sin^2\theta) = 0.$

$\Rightarrow \sin^2\theta - 4\sin\theta - 1 = 0 \Rightarrow \sin\theta = 2 \pm \sqrt{5}.$

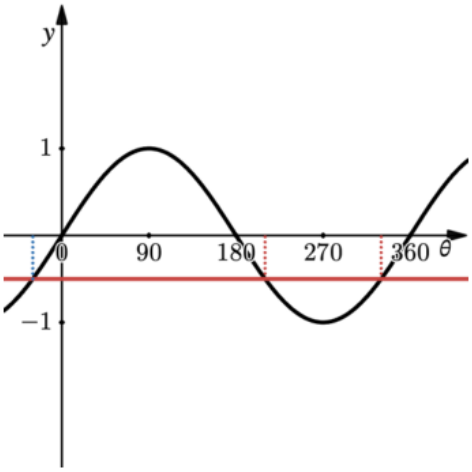
As for $\sin\theta = 2 + \sqrt{5}$ there are no solutions.



Now solve $\sin\theta = 2 - \sqrt{5}$ for $0° \leq \theta < 360°$.

Find first value:

$\theta = \arcsin(2 - \sqrt{5}) = -13.654\ldots°$ (in range).



Find first value between 0° and 360°:

$\theta = 180° + 13.654\ldots° = 193.654\ldots°$ (in range).

Find second value between 0° and 360°:

$\theta = 360° - 13.654\ldots° = 346.345\ldots°$ (in range).

$\theta = \{194°, 346°\}$ (3 SF).

26 b)

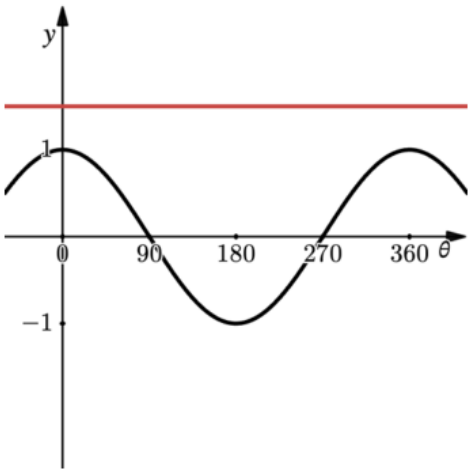
$\sin\theta\tan\theta + 6 = 0$ for $0° \leq \theta < 360°$.

$\Rightarrow \sin\theta \times \frac{\sin\theta}{\cos\theta} + 6 = 0 \Rightarrow \sin^2\theta + 6\cos\theta = 0.$

$\Rightarrow (1 - \cos^2\theta) + 6\cos\theta = 0.$

$\Rightarrow \cos^2\theta - 6\cos\theta - 1 = 0 \Rightarrow \cos\theta = 3 \pm \sqrt{10}.$

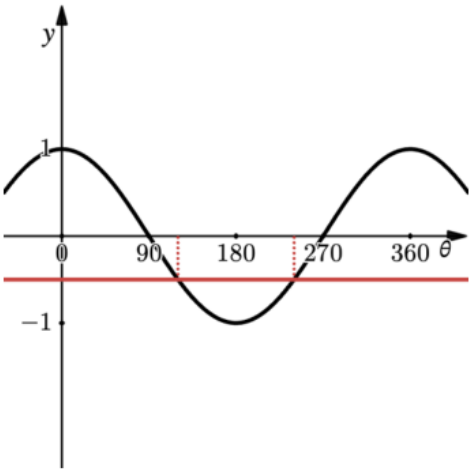
As for $\cos\theta = 3 + \sqrt{10}$ there are no solutions.



Now solve $\cos\theta = 3 - \sqrt{10}$ for $0° \leq \theta < 360°$.

Find first value between 0° and 360°:

$\theta = \arccos(3 - \sqrt{10}) = 99.339\ldots°$ (in range).



Find second value between 0° and 360°:

$\theta = 360° - 99.339\ldots° = 260.660\ldots°$ (in range).

$\Rightarrow \theta = \{99.3°, 261°\}$ (3 SF).

26 c)

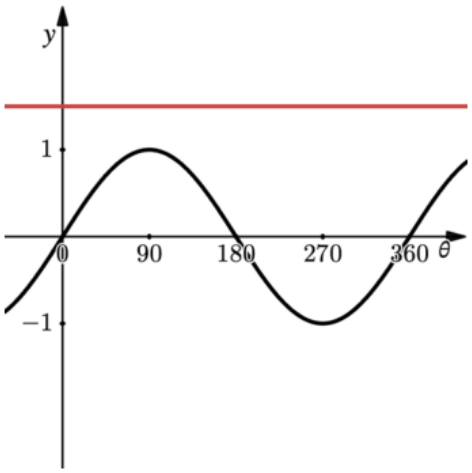
$8\tan 2\theta + \cos 2\theta = 0$ for $0° \leq \theta < 360°$.

$\Rightarrow \frac{8\sin 2\theta}{\cos 2\theta} + \cos 2\theta = 0 \Rightarrow 8\sin 2\theta + \cos^2 2\theta = 0.$

$\Rightarrow 8\sin 2\theta + (1 - \sin^2 2\theta) = 0.$

$\Rightarrow \sin^2 2\theta - 8\sin 2\theta - 1 = 0 \Rightarrow \sin 2\theta = 4 \pm \sqrt{17}.$

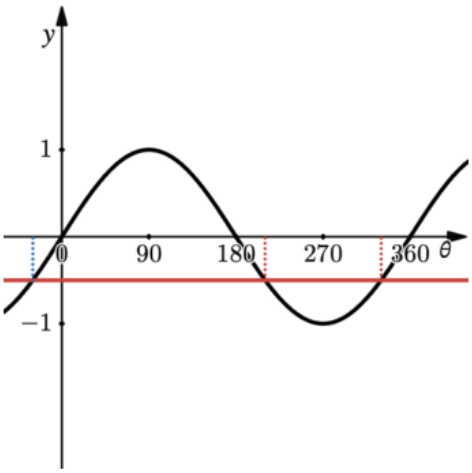
As for $\sin 2\theta = 4 + \sqrt{17}$ there are no solutions.



Now solve $\sin 2\theta = 4 - \sqrt{17}$ for $0° \leq \theta < 360°$.

Find first value:

$2\theta = \arcsin(4 - \sqrt{17}) = -7.071\ldots°.$



Find first value between 0° and 360°:

$2\theta = 180° + 7.071\ldots° = 187.071\ldots°.$

Find second value between 0° and 360°:

$2\theta = 360° - 7.071\ldots° = 352.928\ldots°.$

Apply transformation:

$2\theta = 187.071\ldots° \Rightarrow \theta = 93.535\ldots°$ (in range).

$2\theta = 352.928\ldots° \Rightarrow \theta = 176.464\ldots°$ (in range).

Add period: $\theta = 93.535\ldots^\circ + 180^\circ = 273.535\ldots^\circ$ (in range).

$\theta = 176.464\ldots^\circ + 180^\circ = 356.464\ldots^\circ$ (in range).

$\theta = \{93.5^\circ, 176^\circ, 274^\circ, 356^\circ\}$ (3 SF).

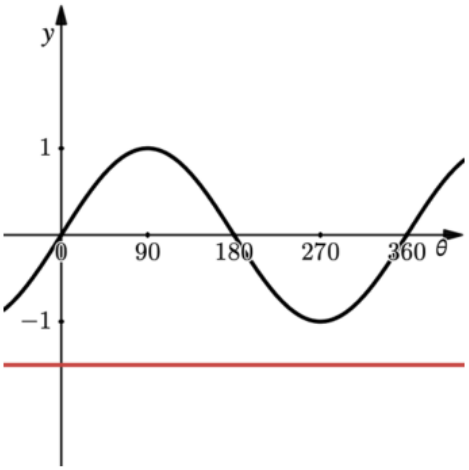
27 a)

$2\cos^2\theta = 3\sin\theta$ for $0^\circ \leq \theta < 180^\circ$.

$\Rightarrow 2(1 - \sin^2\theta) = 3\sin\theta \Rightarrow 2 - 2\sin^2\theta = 3\sin\theta$.

$\Rightarrow 2\sin^2\theta + 3\sin\theta - 2 = 0 \Rightarrow \sin\theta = -2$ or $\sin\theta = \frac{1}{2}$.

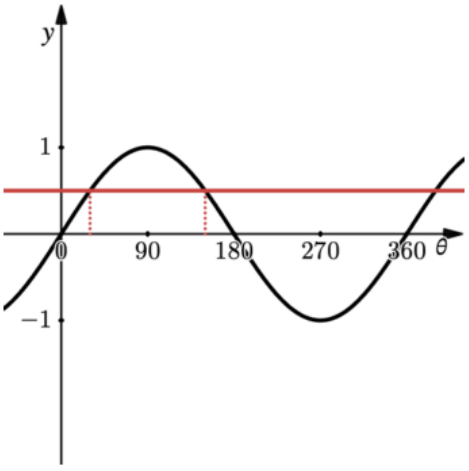
As for $\sin\theta = -2$ there are no solutions.



Now solve $\sin\theta = \frac{1}{2}$ for $0^\circ \leq \theta < 180^\circ$.

Find first value between 0° and 360°:

$\theta = \arcsin\frac{1}{2} = 30^\circ$ (in range).



Find second value between 0° and 360°:

$\theta = 180^\circ - 30^\circ = 150^\circ$ (in range).

$\theta = \{30^\circ, 150^\circ\}$.

27 b)

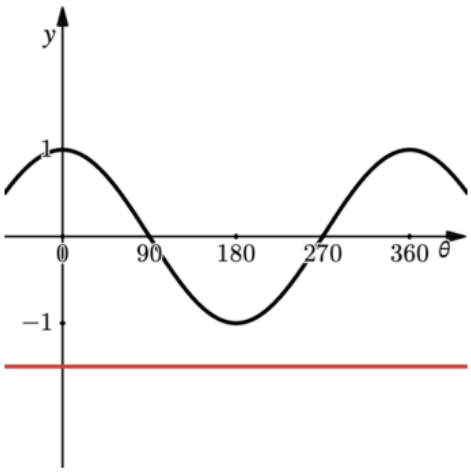
$4\sin^2\theta = 15\cos\theta$ for $0° \leq \theta < 540°$.

$\Rightarrow 4(1 - \cos^2\theta) = 15\cos\theta$.

$\Rightarrow 4 - 4\cos^2\theta = 15\cos\theta$.

$\Rightarrow 4\cos^2\theta + 15\cos\theta - 4 = 0 \Rightarrow \cos\theta = -4$ or $\cos\theta = \frac{1}{4}$.

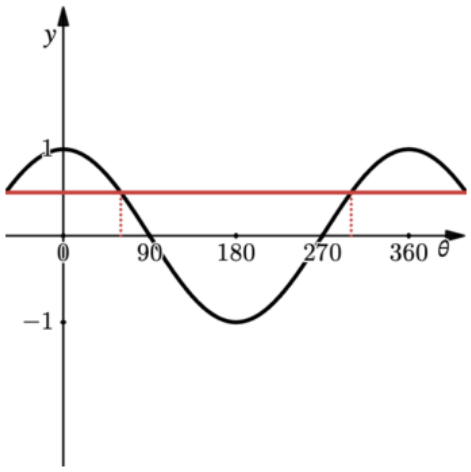
As for $\cos\theta = -4$ there are no solutions.



Now solve $\cos\theta = \frac{1}{4}$ for $0° \leq \theta < 540°$.

Find first value between 0° and 360°:

$\theta = \arccos\frac{1}{4} = 75.522\ldots°$ (in range).



Find second value between 0° and 360°:

$\theta = 360° - 75.522\ldots° = 284.477\ldots°$ (in range).

Add period:

$\theta = 75.522\ldots° + 360° = 435.522\ldots°$ (in range).

$\Rightarrow \theta = \{75.5°, 284°, 436°\}$ (3 SF).

27 c)

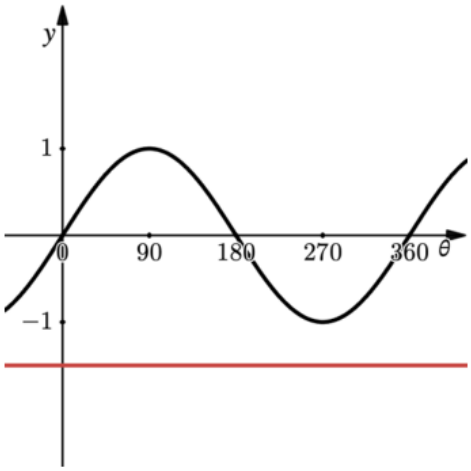
$7\cos^2\theta = 48\sin\theta$ for $0° \leq \theta < 360°$.

$\Rightarrow 7(1-\sin^2\theta) = 48\sin\theta$.

$\Rightarrow 7 - 7\sin^2\theta = 48\sin\theta$.

$\Rightarrow 7\sin^2\theta + 48\sin\theta - 7 = 0 \Rightarrow \sin\theta = -7$ or $\sin\theta = \frac{1}{7}$.

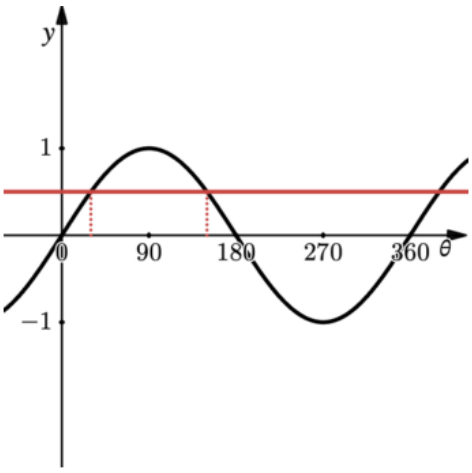
As for $\sin\theta = -7$ there are no solutions.



Now solve $\sin\theta = \frac{1}{7}$ for $0° \leq \theta < 360°$.

Find first value between 0° and 360°:

$\theta = \arcsin\frac{1}{7} = 8.213\ldots°$ (in range).



Find second value between 0° and 360°:

$\theta = 180° - 8.213\ldots° = 171.786\ldots°$ (in range).

$\theta = \{8.21°, 172°\}$ (3 SF).

27 d)

$10\sin^2\theta = 99\cos\theta$ for $-180° \leq \theta < 360°$.

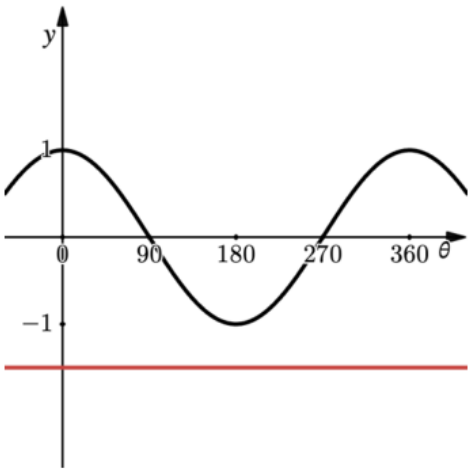
$\Rightarrow 10(1-\cos^2\theta) = 99\cos\theta$.

$\Rightarrow 10 - 10\cos^2\theta = 99\cos\theta$.

$\Rightarrow 10\cos^2\theta + 99\cos\theta - 10 = 0$.

$\Rightarrow \cos\theta = -10$ or $\cos\theta = \frac{1}{10}$.

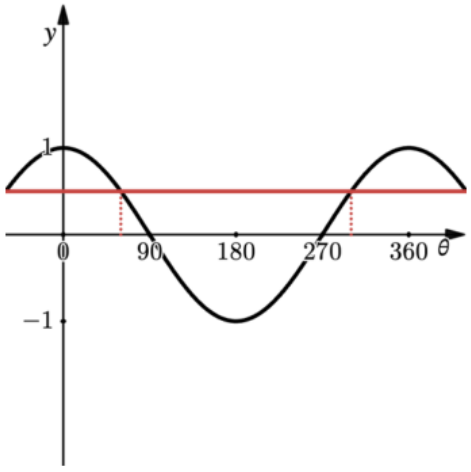
As for $\cos\theta = -10$ there are no solutions.



Now solve $\cos\theta = \frac{1}{10}$ for $-180° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

$\theta = \arccos\frac{1}{10} = 84.260\ldots°$ (in range).



Find second value between $0°$ and $360°$:

$\theta = 360° - 84.260\ldots° = 275.739\ldots°$ (in range).

Subtract period:

$\theta = 275.739\ldots° - 360° = -84.260\ldots°$ (in range).

$\Rightarrow \theta = \{-84.3°, 84.3°, 276°\}$ (3 SF).

28)

$$\begin{aligned} 2\cos^4 x + \cos^2 x &= 2(\cos^2 x)^2 + \cos^2 x \\ &= 2(1-\sin^2 x)^2 + (1-\sin^2 x) \\ &= 2(1-2\sin^2 x + \sin^4 x) + 1 - \sin^2 x \\ &= 2 - 4\sin^2 x + 2\sin^4 x + 1 - \sin^2 x \\ &= 2\sin^4 x - 5\sin^2 x + 3. \end{aligned}$$

29 a)

$2\cos^2\theta + 3\sin\theta - 3 = 0$ for $0° \leq \theta < 360°$.

$\Rightarrow 2(1-\sin^2\theta) + 3\sin\theta - 3 = 0.$

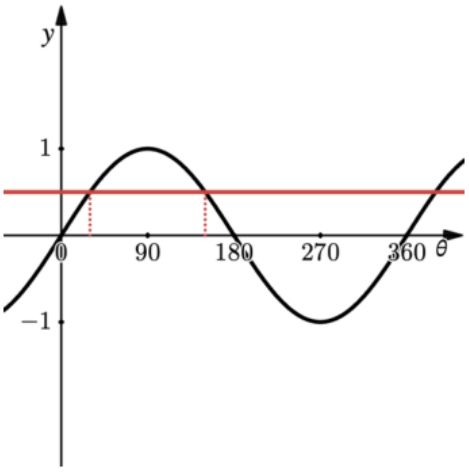
$\Rightarrow 2 - 2\sin^2\theta + 3\sin\theta - 3 = 0.$

$\Rightarrow 2\sin^2\theta - 3\sin\theta + 1 = 0 \Rightarrow \sin\theta = \frac{1}{2}$ or $\sin\theta = 1$.

First solve $\sin\theta = \frac{1}{2}$ for $0° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

$\theta = \arcsin\frac{1}{2} = 30°$ (in range).

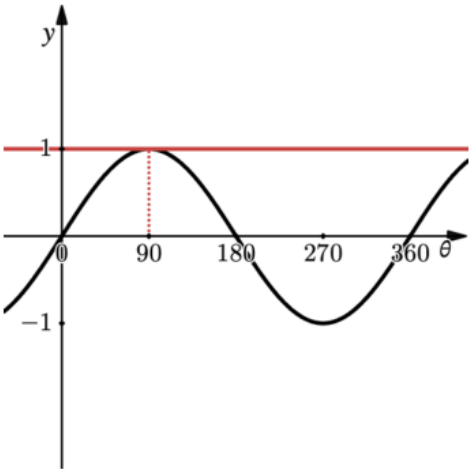


Find second value between $0°$ and $360°$:

$\theta = 180° - 30° = 150°$ (in range).

Now solve $\sin\theta = 1$ for $0° \leq \theta < 360°$.

$\theta = \arcsin 1 = 90°$.



$\theta = \{30°, 90°, 150°\}$.

29 b)

$3\sin^2\theta - 5\cos\theta - 5 = 0$ for $0° \leq \theta < 360°$.

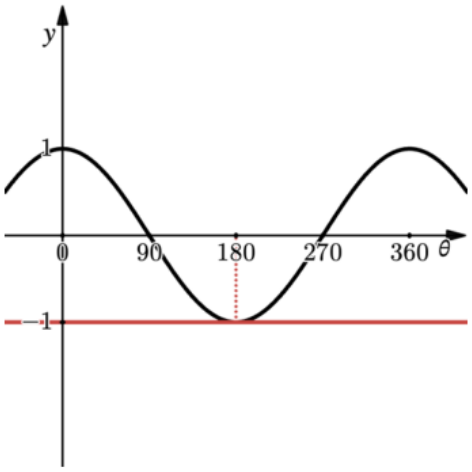
$\Rightarrow 3(1 - \cos^2\theta) - 5\cos\theta - 5 = 0.$

$\Rightarrow 3 - 3\cos^2\theta - 5\cos\theta - 5 = 0.$

$\Rightarrow 3\cos^2\theta + 5\cos\theta + 2 = 0 \Rightarrow \cos\theta = -1$ or $\cos\theta = -\frac{2}{3}$.

First solve $\cos\theta = -1$ for $0° \leq \theta < 360°$.

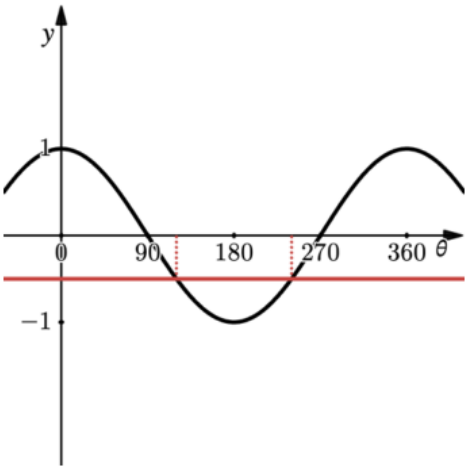
$\theta = \arccos(-1) = 180°$.



Now solve $\cos\theta = -\frac{2}{3}$ for $0° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

$\theta = \arccos(-\frac{2}{3}) = 131.810\ldots°$ (in range).



Find second value between 0° and 360°:

$\theta = 360° - 131.810\ldots° = 228.189\ldots°$ (in range).

$\Rightarrow \theta = \{132°, 180°, 228°\}$ (3 SF).

29 c)

$6\cos^2\theta + \sin\theta - 4 = 0$ for $0° \leq \theta < 360°$.

$\Rightarrow 6(1 - \sin^2\theta) + \sin\theta - 4 = 0.$

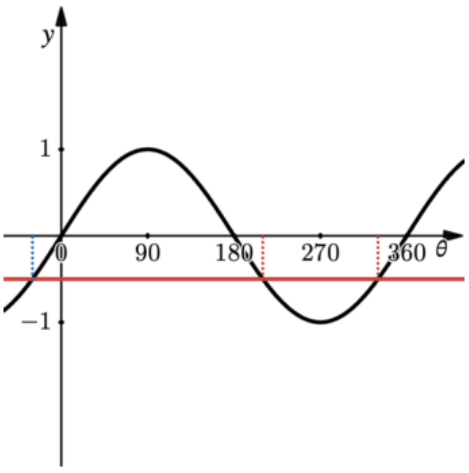
$\Rightarrow 6 - 6\sin^2\theta + \sin\theta - 4 = 0.$

$\Rightarrow 6\sin^2\theta - \sin\theta - 2 = 0 \Rightarrow \sin\theta = -\frac{1}{2}$ or $\sin x = \frac{2}{3}$.

First solve $\sin\theta = -\frac{1}{2}$ for $0° \leq \theta < 360°$.

Find first value:

$\theta = \arcsin(-\frac{1}{2}) = -30°$ (in range).



Find first value between 0° and 360°:

$\theta = 180° + 30° = 210°$ (in range).

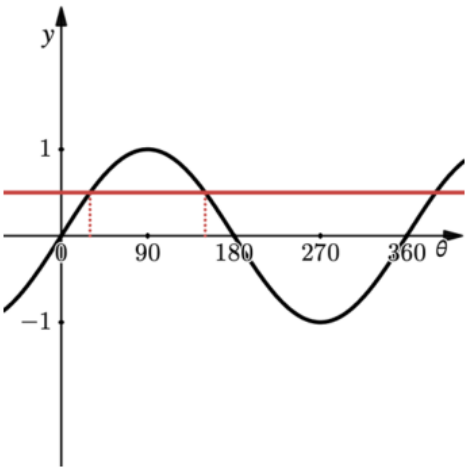
Find second value between 0° and 360°:

$\theta = 360° - 30° = 330°$ (in range).

Now solve $\sin\theta = \frac{2}{3}$ for $0° \leq \theta < 360°$.

Find first value between 0° and 360°:

$\theta = \arcsin\frac{2}{3} = 41.810\ldots°$ (in range).



Find second value between 0° and 360°:

$\theta = 180° - 41.810\ldots° = 138.189\ldots°$ (in range).

$\theta = \{41.8°, 138°, 210°, 330°\}$ (3 SF).

29 d)

$20\sin^2\theta - 7\cos\theta - 14 = 0$ for $0° \leq \theta < 360°$.

$\Rightarrow 20(1 - \cos^2\theta) - 7\cos\theta - 14 = 0.$

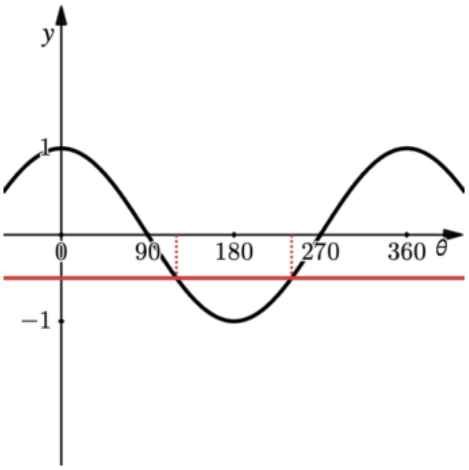
$\Rightarrow 20 - 20\cos^2\theta - 7\cos\theta - 14 = 0.$

$\Rightarrow 20\cos^2\theta + 7\cos\theta - 6 = 0 \Rightarrow \cos\theta = -\frac{3}{4}$ or $\cos\theta = \frac{2}{5}$.

First solve $\cos\theta = -\frac{3}{4}$ for $0° \leq \theta < 360°$.

Find first value between 0° and 360°:

$\theta = \arccos(-\frac{3}{4}) = 138.590\ldots°$ (in range).



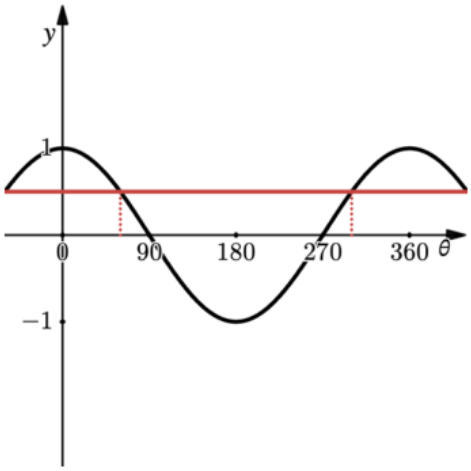
Find second value between 0° and 360°:

$\theta = 360° - 138.590\ldots° = 221.409\ldots°$ (in range).

Now solve $\cos\theta = \frac{2}{5}$ for $0° \leq \theta < 360°$.

Find first value between 0° and 360°:

$\theta = \arccos \frac{2}{5} = 66.421\ldots°$ (in range).



Find second value between 0° and 360°:

$\theta = 360° - 66.421\ldots° = 293.578\ldots°$ (in range).

$\Rightarrow \theta = \{66.4°, 139°, 221°, 294°\}$ (3 SF).

30 a)

$3\sin^2\theta + \sin\theta - 3\cos^2\theta + 2 = 0$ for $0° \leq \theta < 360°$.

$\Rightarrow 3\sin^2\theta + \sin\theta - 3(1 - \sin^2\theta) + 2 = 0.$

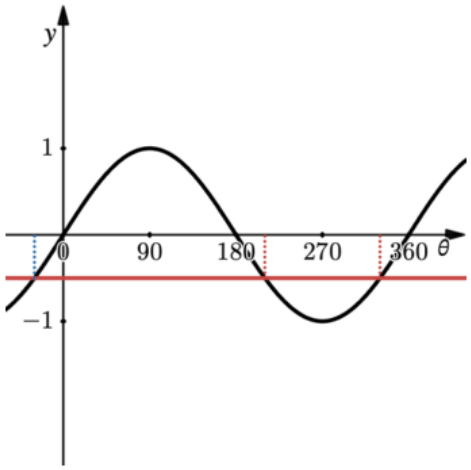
$\Rightarrow 3\sin^2\theta + \sin\theta - 3 + 3\sin^2\theta + 2 = 0.$

$\Rightarrow 6\sin^2\theta + \sin\theta - 1 = 0 \Rightarrow \sin\theta = -\frac{1}{2}$ or $\sin\theta = \frac{1}{3}$.

First solve $\sin\theta = -\frac{1}{2}$ for $0° \leq \theta < 360°$.

Find first value:

$\theta = \arcsin(-\frac{1}{2}) = -30°$ (in range).



Find first value between 0° and 360°:

$\theta = 180° + 30° = 210°$ (in range).

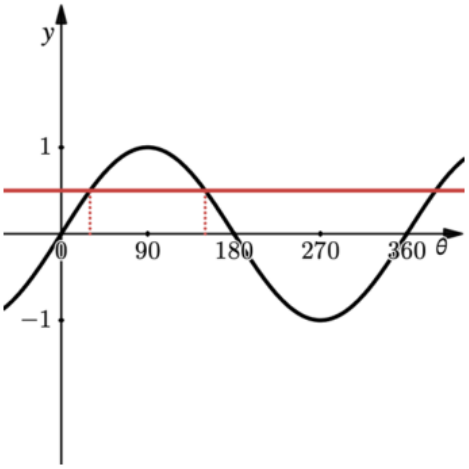
Find second value between 0° and 360°:

$\theta = 360° - 30° = 330°$ (in range).

Now solve $\sin\theta = \frac{1}{3}$ for $0° \leq \theta < 360°$.

Find first value between 0° and 360°:

$\theta = \arcsin\frac{1}{3} = 19.471\ldots°$ (in range).



Find second value between 0° and 360°:

$\theta = 180° - 19.471\ldots° = 160.528\ldots°$ (in range).

$\theta = \{19.5°, 161°, 210°, 330°\}$ (3 SF).

30 b)

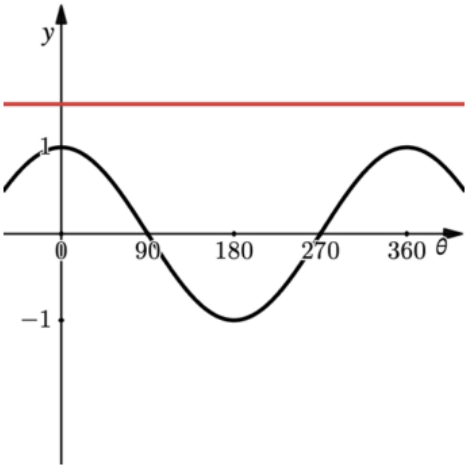
$4\cos^2\theta - 16\cos\theta + 3\sin^2\theta - 19 = 0$ for $0° \leq \theta < 360°$.

$\Rightarrow 4\cos^2\theta - 16\cos\theta + 3(1 - \cos^2\theta) - 19 = 0.$

$\Rightarrow 4\cos^2\theta - 16\cos\theta + 3 - 3\cos^2\theta - 19 = 0.$

$\Rightarrow \cos^2\theta - 16\cos\theta - 16 = 0 \Rightarrow \cos\theta = 8 \pm 4\sqrt{5}.$

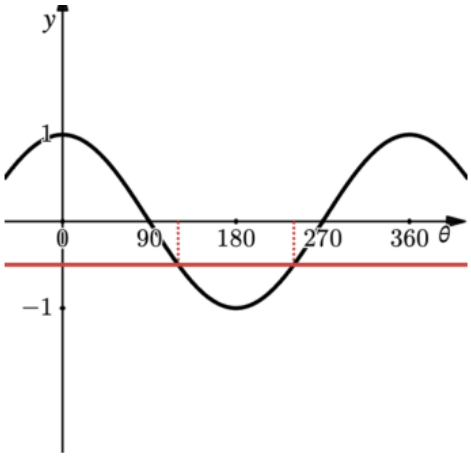
As for $\cos\theta = 8 + 4\sqrt{5}$, there are no solutions:



Now solve $\cos\theta = 8 - 4\sqrt{5}$ for $0° \leq \theta < 360°$.

Find first value between 0° and 360°:

$\theta = \arccos(8 - 4\sqrt{5}) = 160.781\ldots°$ (in range).



0° and 360°:

Find second value between 0° and 360°:

$\theta = 360° - 160.781\ldots° = 199.218\ldots°$ (in range).

$\Rightarrow \theta = \{161°, 199°\}$ (3 SF).

30 c)

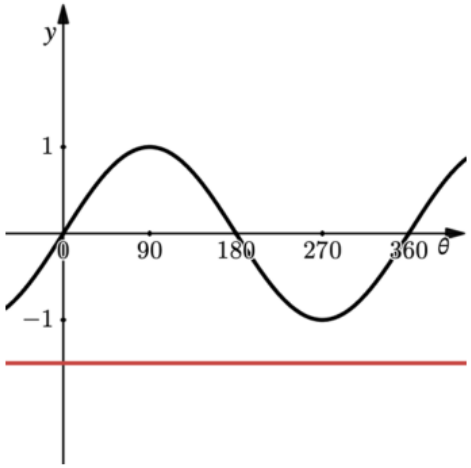
$10\sin^2\theta - 11\cos^2\theta + 46\sin\theta + 35 = 0$ for $0° \leq \theta < 360°$.

$\Rightarrow 10\sin^2\theta - 11(1 - \sin^2\theta) + 46\sin\theta + 35 = 0.$

$\Rightarrow 10\sin^2\theta - 11 + 11\sin^2\theta + 46\sin\theta + 35 = 0.$

$\Rightarrow 21\sin^2\theta + 46\sin\theta + 24 = 0 \Rightarrow \sin\theta = -\frac{4}{3}$ or $\sin\theta = -\frac{6}{7}$.

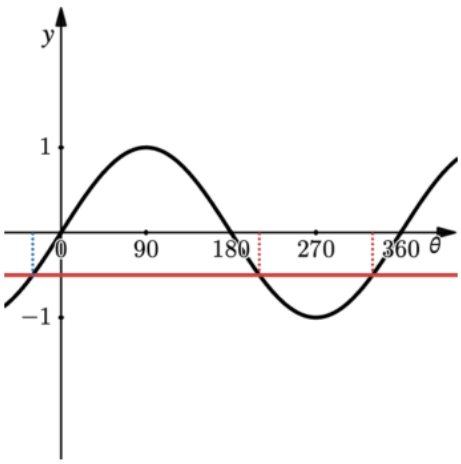
As for $\sin\theta = -\frac{4}{3}$ there are no solutions:



Now solve $\sin\theta = -\frac{6}{7}$ for $0° \leq \theta < 360°$.

Find first value:

$\theta = \arcsin(-\frac{6}{7}) = -58.997\ldots°$ (in range).



Find first value between 0° and 360°:

$\theta = 180° + 58.997\ldots° = 238.997\ldots°$ (in range).

Find second value between 0° and 360°:

$\theta = 360° - 58.997\ldots° = 301.002\ldots°$ (in range).

$\Rightarrow \theta = \{239°, 301°\}$ (3 SF).

31 a)

$5\cos^2 x + 3\cos x + 2 = 4\sin^2 x$ for $-360° \leq x < 360°$.

$\Rightarrow 5\cos^2 x + 3\cos x + 2 = 4(1 - \cos^2 x)$.

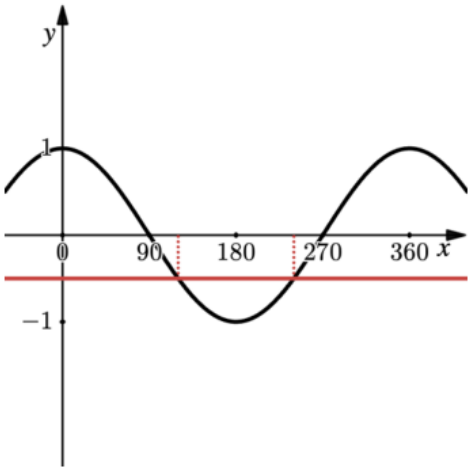
$\Rightarrow 5\cos^2 x + 3\cos x + 2 = 4 - 4\cos^2 x$.

$\Rightarrow 9\cos^2 x + 3\cos x - 2 = 0 \Rightarrow \cos x = -\frac{2}{3}$ or $\cos x = \frac{1}{3}$.

First solve $\cos x = -\frac{2}{3}$ for $0° \leq x < 360°$.

Find first value between 0° and 360°:

$x = \arccos(-\frac{2}{3}) = 131.810\ldots°$ (in range).



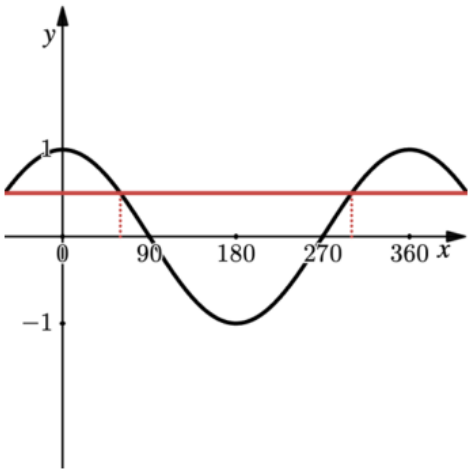
Find second value between 0° and 360°:

$x = 360° - 131.810\ldots° = 228.189\ldots°$ (in range).

Now solve $\cos x = \frac{1}{3}$ for $0° \leq x < 360°$.

Find first value between 0° and 360°:

$x = \arccos \frac{1}{3} = 70.528\ldots°$ (in range).



Find second value between 0° and 360°:

$x = 360° - 70.528\ldots° = 289.471\ldots°$ (in range).

$\Rightarrow x = \{70.5°, 132°, 228°, 289°\}$ (3 SF).

31 b)

$2\theta + 50° = 70.528\ldots \Rightarrow \theta = 10.264\ldots° = 10.3°$ (3 SF).

32)

$2\sin^2 x + 3 = \tan 5x - 2\cos^2 x$ for $0° \leq x < 72°$.

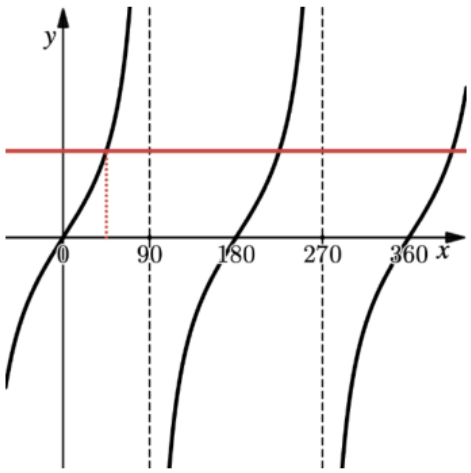
$\Rightarrow 2\sin^2 x + 2\cos^2 x + 3 = \tan 5x.$

$\Rightarrow 2(\sin^2 x + \cos^2 x) + 3 = \tan 5x.$

$\Rightarrow 2 \times 1 + 3 = \tan 5x \Rightarrow \tan 5x = 5.$

Find first value:

$5x = \arctan 5 = 78.690\ldots°.$



Apply transformation:

$5x = 78.690\ldots^\circ \Rightarrow x = 15.738\ldots^\circ$ (in range).

Add period:

$\theta = 15.738\ldots^\circ + 36^\circ = 51.738\ldots^\circ$ (in range).

$\theta = \{15.7^\circ, 51.7^\circ\}$ (3 SF).

33 a)

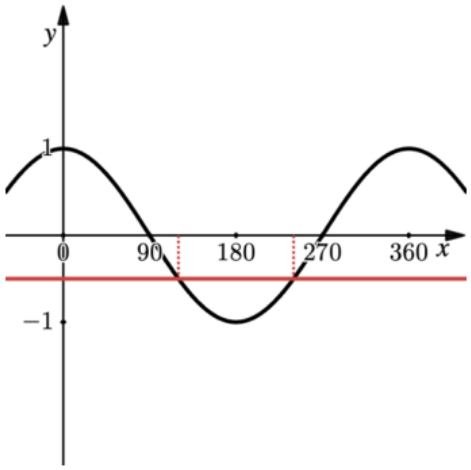
$10\sin\theta\tan\theta + 21 = 0$ for $0^\circ \leq \theta < 360^\circ$.

$$\begin{aligned}
&\Rightarrow 10\sin\theta \times \frac{\sin\theta}{\cos\theta} + 21 = 0 \\
&\Rightarrow 10\sin^2\theta + 21\cos\theta = 0 \\
&\Rightarrow 10(1 - \cos^2\theta) + 21\cos\theta = 0 \\
&\Rightarrow 10 - 10\cos^2\theta + 21\cos\theta = 0 \\
&\Rightarrow 10\cos^2\theta - 21\cos\theta - 10 = 0 \\
&\Rightarrow \cos\theta = -\frac{2}{5} \text{ or } \cos\theta = \frac{5}{2}.
\end{aligned}$$

First solve $\cos\theta = -\frac{2}{5}$ for $0^\circ \leq \theta < 360^\circ$.

Find first value between 0° and 360°:

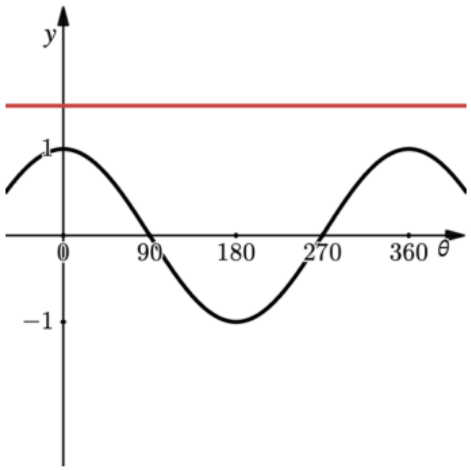
$x = \arccos(-\frac{2}{5}) = 113.578\ldots^\circ$ (in range).



Find second value between 0° and 360°:

$x = 360^\circ - 113.578\ldots^\circ = 246.421\ldots^\circ$ (in range).

As for $\cos\theta = \frac{5}{2}$, there are no solutions:



$\Rightarrow \theta = \{114°, 246°\}$ (3 SF).

33 b)

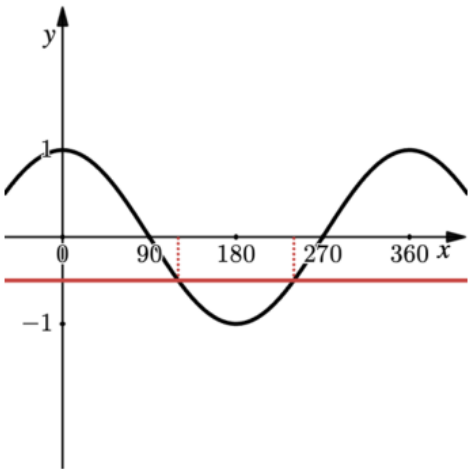
$2\sin\theta\tan\theta + 20\cos\theta + 15 = 0$ for $0° \leq \theta < 360°$.

$$\Rightarrow 2\sin\theta \times \frac{\sin\theta}{\cos\theta} + 20\cos\theta + 15 = 0$$
$$\Rightarrow 2\sin^2\theta + 20\cos^2\theta + 15\cos\theta = 0$$
$$\Rightarrow 2(1 - \cos^2\theta) + 20\cos^2\theta + 15\cos\theta = 0$$
$$\Rightarrow 2 - 2\cos^2\theta + 20\cos^2\theta + 15\cos\theta = 0$$
$$\Rightarrow 18\cos^2\theta + 15\cos\theta + 2 = 0 \Rightarrow \cos\theta = -\frac{2}{3} \text{ or } \cos\theta = -\frac{1}{6}.$$

First solve $\cos\theta = -\frac{2}{3}$ for $0° \leq \theta < 360°$.

Find first value between 0° and 360°:

$x = \arccos(-\frac{2}{3}) = 131.810\ldots°$ (in range).



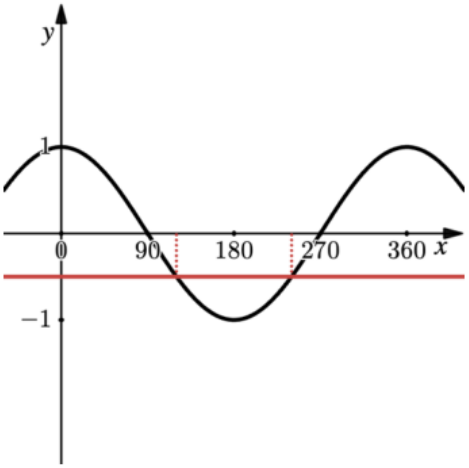
Find second value between 0° and 360°:

$x = 360° - 131.810\ldots° = 228.189\ldots°$ (in range).

Now solve $\cos\theta = -\frac{1}{6}$ for $0° \leq \theta < 360°$.

Find first value between 0° and 360°:

$x = \arccos(-\frac{1}{6}) = 99.594\ldots°$ (in range).



Find second value between 0° and 360°:

$x = 360° - 99.594\ldots° = 260.405\ldots°$ (in range).

$\Rightarrow \theta = \{99.6°, 132°, 228°, 260°\}$ (3 SF).

33 c)

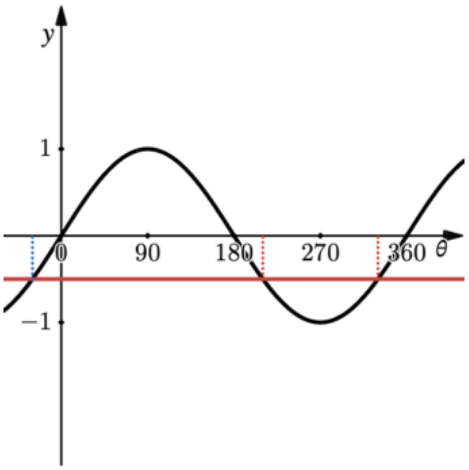
$\dfrac{2\cos\theta}{\tan\theta} = 18\sin\theta - 3$ for $0° \leq \theta < 360°$.

$$\Rightarrow 2\cos\theta = 18\sin\theta\tan\theta - 3\tan\theta$$
$$\Rightarrow 2\cos\theta = 18\sin\theta \times \frac{\sin\theta}{\cos\theta} - \frac{3\sin\theta}{\cos\theta}$$
$$\Rightarrow 2\cos^2\theta = 18\sin^2\theta - 3\sin\theta$$
$$\Rightarrow 2(1 - \sin^2\theta) = 18\sin^2\theta - 3\sin\theta$$
$$\Rightarrow 2 - 2\sin^2\theta = 18\sin^2\theta - 3\sin\theta$$
$$\Rightarrow 20\sin^2\theta - 3\sin\theta - 2 = 0 \Rightarrow \sin\theta = -\frac{1}{4} \text{ or } \sin\theta = \frac{2}{5}.$$

First solve $\sin\theta = -\frac{1}{4}$ for $0° \leq \theta < 360°$.

Find first value:

$\theta = \arcsin(-\frac{1}{4}) = -14.477\ldots°$ (in range).



Find first value between 0° and 360°:

$\theta = 180° + 14.477\ldots° = 194.477\ldots°$ (in range).

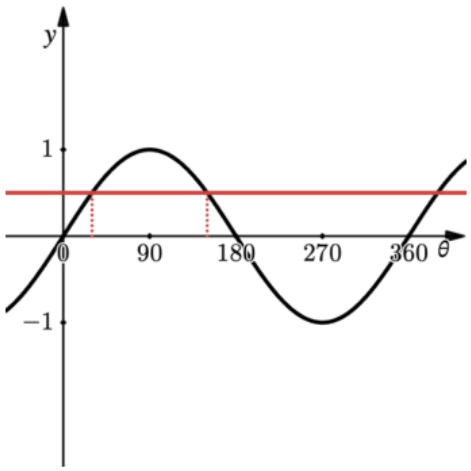
Find second value between 0° and 360°:

$\theta = 360° - 14.477\ldots° = 345.522\ldots°$ (in range).

Now solve $\sin\theta = \frac{2}{5}$ for $0° \leq \theta < 360°$.

Find first value between 0° and 360°:

$\theta = \arcsin\frac{2}{5} = 23.578\ldots°$ (in range).



Find second value between 0° and 360°:

$\theta = 180° - 23.578\ldots° = 156.421\ldots°$ (in range).

$\Rightarrow \theta = \{23.6°, 156°, 194°, 346°\}$ (3 SF).

34)

$12\sin^2 x - 28\sin x\cos x = 5\cos^2 x.$

$\Rightarrow 12\sin^2 x - 28\sin x\cos x - 5\cos^2 x = 0.$

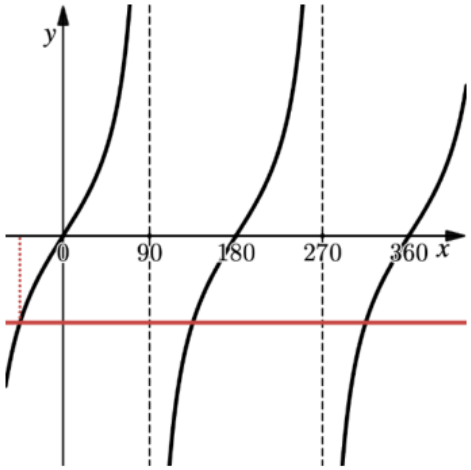
$\Rightarrow (6\sin x + \cos x)(2\sin x - 5\cos x) = 0.$

First solve $6\sin x + \cos x = 0$ for $0° \leq x < 360°$.

$\Rightarrow 6\sin x = -\cos x \Rightarrow \dfrac{\sin x}{\cos x} = -\frac{1}{6} \Rightarrow \tan x = -\frac{1}{6}.$

Find first value:

$x = \arctan(-\frac{1}{6}) = -9.462\ldots°$ (not in range).



Add on period:

$x = -9.642\ldots° + 180° = 170.537\ldots°$ (in range).

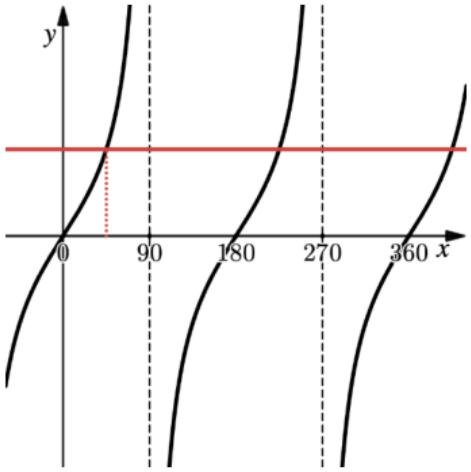
$x = 170.537\ldots° + 180° = 350.537\ldots°$ (in range).

Now solve $2\sin x - 5\cos x = 0$ for $0° \leq x < 360°$.

$\Rightarrow 2\sin x = 5\cos x \Rightarrow \dfrac{\sin x}{\cos x} = \frac{5}{2} \Rightarrow \tan x = \frac{5}{2}.$

Find first value:

$x = \arctan\frac{5}{2} = 68.198\ldots°$ (in range).

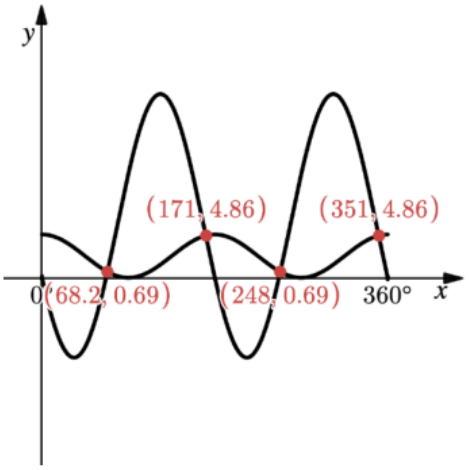


Add period: $x = 68.198\ldots° + 180° = 248.198\ldots°$ (in range).

$\Rightarrow x = \{68.2°, 171°, 248°, 351°\}$ (3 SF).

Substitute these x-values into $y = 5\cos^2 x$ to find the y-coordinates:

$(68.2°, 0.690)$, $(171°, 4.86)$, $(248°, 0.690)$, $(351°, 4.86)$ (3 SF).



35 a)

$\sin\theta\tan\theta + 4\cos\theta = 5\sin\theta$ for $0° \leq \theta < 360°$.

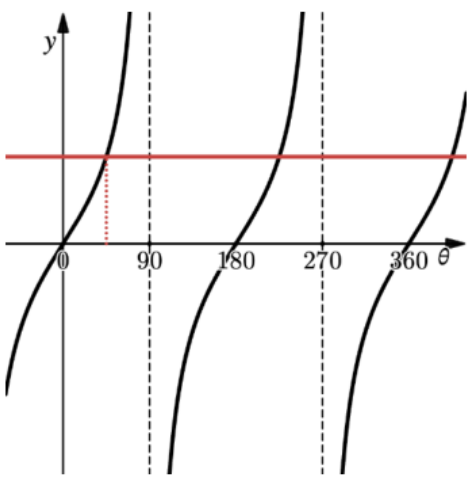
$$\Rightarrow \sin\theta \times \frac{\sin\theta}{\cos\theta} + 4\cos\theta = 5\sin\theta$$
$$\Rightarrow \sin^2\theta + 4\cos^2\theta = 5\sin\theta\cos\theta$$
$$\Rightarrow \sin^2\theta - 5\sin\theta\cos\theta + 4\cos^2\theta = 0$$
$$\Rightarrow (\sin\theta - \cos\theta)(\sin\theta - 4\cos\theta) = 0.$$

First solve $\sin\theta - \cos\theta = 0$ for $0° \leq \theta < 360°$.

$\Rightarrow \dfrac{\sin\theta}{\cos\theta} - 1 = 0 \Rightarrow \tan\theta - 1 = 0 \Rightarrow \tan\theta = 1.$

Find first value:

$\theta = \arctan 1 = 45°$ (in range).



Add period:

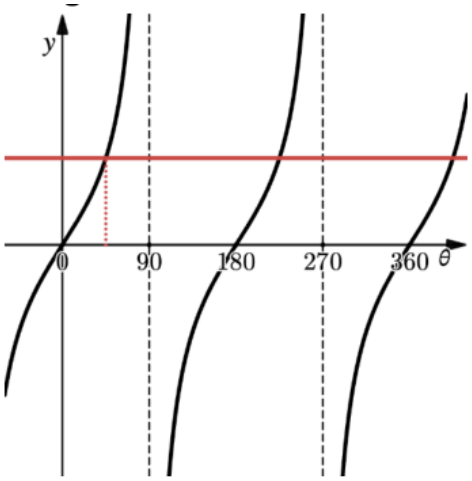
$\theta = 45° + 180° = 225°$ (in range).

Now solve $\sin\theta - 4\cos\theta = 0$ for $0° \leq \theta < 360°$.

$\Rightarrow \dfrac{\sin\theta}{\cos\theta} - 4 = 0 \Rightarrow \tan\theta - 4 = 0 \Rightarrow \tan\theta = 4.$

Find first value:

$\theta = \arctan 4 = 75.963\ldots°$ (in range).



Add period:

$\theta = 75.963\ldots° + 180° = 255.963\ldots°$ (in range).

$\Rightarrow \theta = \{45°, 76.0°, 225°, 256°\}$ (3 SF).

35 b)

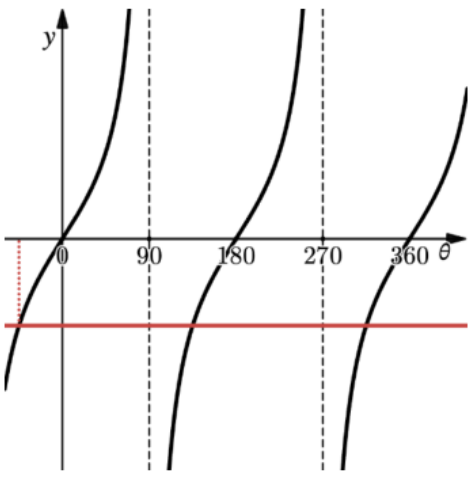
$5\sin\theta\tan\theta = 3\cos\theta + 14\sin\theta$ for $0° \leq \theta < 360°$.

$$\Rightarrow 5\sin\theta \times \frac{\sin\theta}{\cos\theta} = 3\cos\theta + 14\sin\theta$$
$$\Rightarrow 5\sin^2\theta = 3\cos^2\theta + 14\sin\theta\cos\theta$$
$$\Rightarrow 5\sin^2\theta - 14\sin\theta\cos\theta - 3\cos^2\theta = 0$$
$$\Rightarrow (5\sin\theta + \cos\theta)(\sin\theta - 3\cos\theta) = 0.$$

First solve $5\sin\theta + \cos\theta = 0 \Rightarrow \dfrac{5\sin\theta}{\cos\theta} + 1 = 0 \Rightarrow 5\tan\theta + 1 = 0 \Rightarrow \tan\theta = -\frac{1}{5}$.

Find first value:

$\theta = \arctan(-\frac{1}{5}) = -11.309\ldots°$ (not in range).



Add on period:

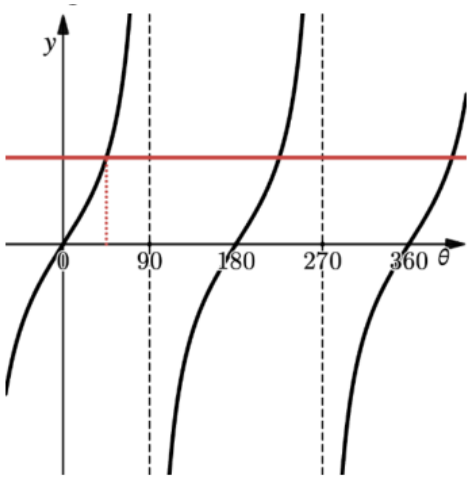
$\theta = -11.309\ldots^\circ + 180^\circ = 168.690\ldots^\circ$ (in range).

$\theta = 168.690\ldots^\circ + 180^\circ = 348.690\ldots^\circ$ (in range).

Now solve $\sin\theta - 3\cos\theta = 0 \Rightarrow \dfrac{\sin\theta}{\cos\theta} - 3 = 0 \Rightarrow \tan\theta - 3 = 0 \Rightarrow \tan\theta = 3$.

Find first value:

$\theta = \arctan 3 = 71.565\ldots^\circ$ (in range).



Add period: $\theta = 71.565\ldots^\circ + 180^\circ = 251.565\ldots^\circ$ (in range).

$\Rightarrow \theta = \{71.6^\circ, 169^\circ, 252^\circ, 349^\circ\}$ (3 SF).

35 c)

$10\sin\theta\tan\theta + 3\sin\theta = 18\cos\theta$ for $0° \leq \theta < 360°$.

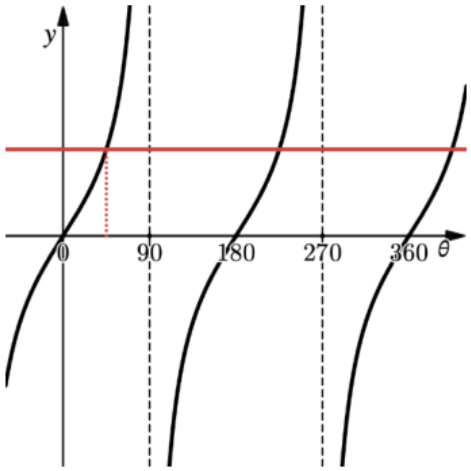
$$\Rightarrow 10\sin\theta \times \frac{\sin\theta}{\cos\theta} + 3\sin\theta = 18\cos\theta$$
$$\Rightarrow 10\sin^2\theta + 3\sin\theta\cos\theta = 18\cos^2\theta$$
$$\Rightarrow 10\sin^2\theta + 3\sin\theta\cos\theta - 18\cos^2\theta = 0$$
$$\Rightarrow (5\sin\theta - 6\cos\theta)(2\sin\theta + 3\cos\theta) = 0.$$

First solve $5\sin\theta - 6\cos\theta = 0$ for $0° \leq \theta < 360°$.

$\Rightarrow \dfrac{5\sin\theta}{\cos\theta} - 6 = 0 \Rightarrow 5\tan\theta = 6 \Rightarrow \tan\theta = \frac{6}{5}$.

Find first value:

$\theta = \arctan\frac{6}{5} = 50.194\ldots°$ (in range).



Add period:

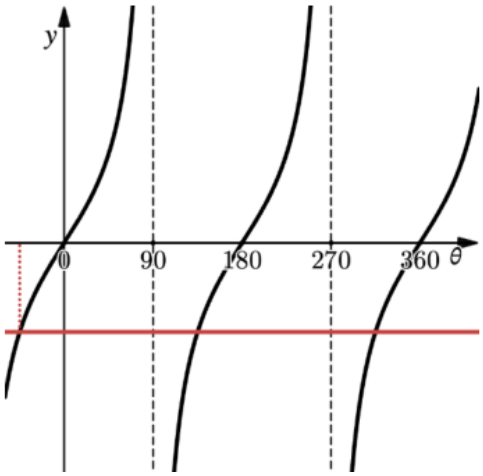
$\theta = 50.194\ldots° + 180° = 230.194\ldots°$ (in range).

Now solve $2\sin\theta + 3\cos\theta = 0$ for $0° \leq \theta < 360°$.

$\Rightarrow \dfrac{2\sin\theta}{\cos\theta} + 3 = 0 \Rightarrow 2\tan\theta + 3 = 0 \Rightarrow \tan\theta = -\frac{3}{2}$.

Find first value:

$\theta = \arctan(-\frac{3}{2}) = -56.309\ldots°$ (not in range).



Add on period:

$\theta = -56.309\ldots^\circ + 180^\circ = 123.690\ldots^\circ$ (in range).

$\theta = 123.690\ldots^\circ + 180^\circ = 303.690\ldots^\circ$ (in range).

$\Rightarrow \theta = \{50.2^\circ, 124^\circ, 230^\circ, 304^\circ\}$ (3 SF).

36 a)

$2\cos\theta + 13\cos\theta\tan\theta + 6\sin\theta\tan\theta = 0$ for $0^\circ \leq \theta < 360^\circ$.

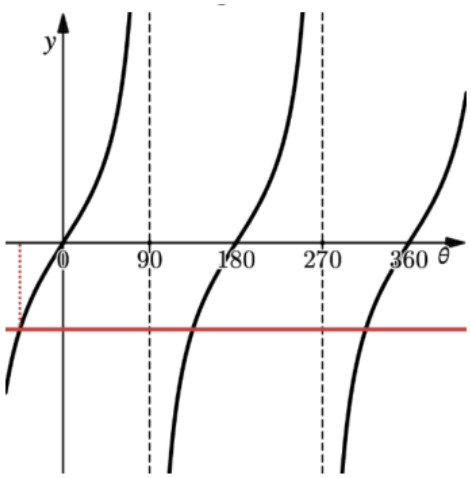
$$\Rightarrow 2\cos\theta + 13\cos\theta \times \frac{\sin\theta}{\cos\theta} + 6\sin\theta \times \frac{\sin\theta}{\cos\theta} = 0$$
$$\Rightarrow 2\cos\theta + 13\sin\theta + \frac{6\sin^2\theta}{\cos\theta} = 0$$
$$\Rightarrow 2\cos^2\theta + 13\sin\theta\cos\theta + 6\sin^2\theta = 0$$
$$\Rightarrow (2\cos\theta + \sin\theta)(\cos\theta + 6\sin\theta) = 0.$$

First solve $2\cos\theta + \sin\theta = 0$ for $0^\circ \leq \theta < 360^\circ$.

$\Rightarrow 2 + \frac{\sin\theta}{\cos\theta} = 0 \Rightarrow 2 + \tan\theta = 0 \Rightarrow \tan\theta = -2.$

Find first value:

$\theta = \arctan(-2) = -63.434\ldots^\circ$ (not in range).



Add on period:

$\theta = -63.434\ldots^\circ + 180^\circ = 116.565\ldots^\circ$ (in range).

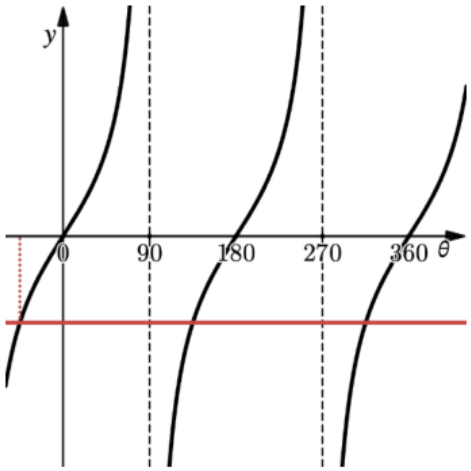
$\theta = 116.565\ldots^\circ + 180^\circ = 296.565\ldots^\circ$ (in range).

Now solve $\cos\theta + 6\sin\theta = 0$ for $0^\circ \leq \theta < 360^\circ$.

$\Rightarrow 1 + \frac{6\sin\theta}{\cos\theta} = 0 \Rightarrow 1 + 6\tan\theta = 0 \Rightarrow \tan\theta = -\frac{1}{6}.$

Find first value:

$\theta = \arctan(-\frac{1}{6}) = -9.462\ldots^\circ$ (not in range).



Add on period:

$\theta = -9.462\ldots^\circ + 180^\circ = 170.537\ldots^\circ$ (in range).

$\theta = 170.537\ldots^\circ + 180^\circ = 350.537\ldots^\circ$ (in range).

$\Rightarrow \theta = \{117^\circ, 171^\circ, 297^\circ, 351^\circ\}$ (3 SF).

36 b)

$15\cos\theta + 6\sin\theta\tan\theta = 19\cos\theta\tan\theta = 0$ for $0^\circ \leq \theta < 360^\circ$.

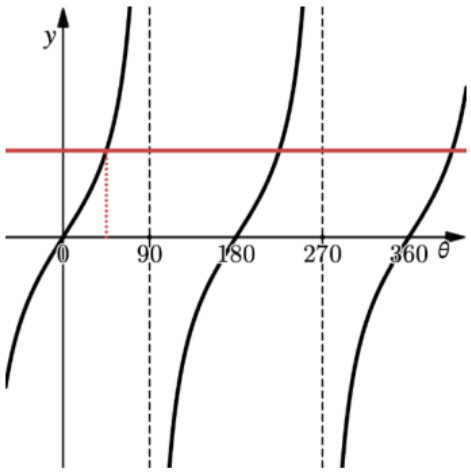
$$\Rightarrow 15\cos\theta + 6\sin\theta \times \frac{\sin\theta}{\cos\theta} = 19\cos\theta \times \frac{\sin\theta}{\cos\theta} = 0$$
$$\Rightarrow 15\cos\theta + \frac{6\sin^2\theta}{\cos\theta} = 19\sin\theta$$
$$\Rightarrow 15\cos^2\theta + 6\sin^2\theta = 19\sin\theta\cos\theta$$
$$\Rightarrow 15\cos^2\theta - 19\sin\theta\cos\theta + 6\sin^2\theta = 0$$
$$\Rightarrow (5\cos\theta - 3\sin\theta)(3\cos\theta - 2\sin\theta) = 0.$$

First solve $5\cos\theta - 3\sin\theta = 0$ for $0^\circ \leq \theta < 360^\circ$.

$\Rightarrow 5 - \dfrac{3\sin\theta}{\cos\theta} = 0 \Rightarrow 5 - 3\tan\theta = 0 \Rightarrow \tan\theta = \frac{5}{3}$.

Find first value:

$\theta = \arctan\frac{5}{3} = 59.036\ldots^\circ$ (in range).



Add period:

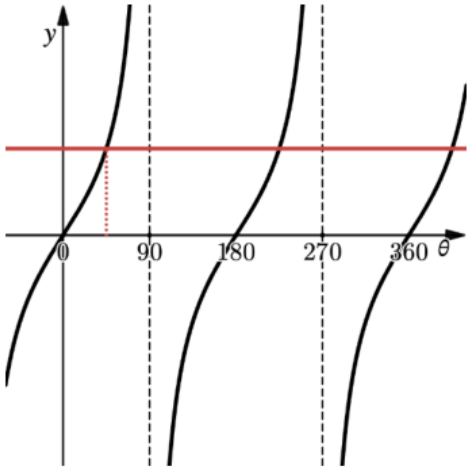
$\theta = 59.036\ldots^\circ + 180^\circ = 239.036\ldots^\circ$ (in range).

Now solve $3\cos\theta - 2\sin\theta = 0$ for $0^\circ \leq \theta < 360^\circ$.

$\Rightarrow 3 - \dfrac{2\sin\theta}{\cos\theta} = 0 \Rightarrow 3 - 2\tan\theta = 0 \Rightarrow \tan\theta = \frac{3}{2}$.

Find first value:

$\theta = \arctan\frac{3}{2} = 56.309\ldots^\circ$ (in range).



Add period:

$\theta = 56.309\ldots^\circ + 180^\circ = 236.309\ldots^\circ$ (in range).

$\Rightarrow \theta = \{56.3^\circ, 59.0^\circ, 236^\circ, 239^\circ\}$ (3 SF).

37 a)

LHS:

$$\begin{aligned}(\sin x + \cos x)^2 &\equiv \sin^2 x + 2\sin x\cos x + \cos^2 x \\ &\equiv (\sin^2 x + \cos^2 x) + 2\sin x\cos x \\ &\equiv 1 + 2\sin x\cos x \quad \text{:RHS.}\end{aligned}$$

37 b)

RHS:

$$\begin{aligned}(\sin x + \cos x)(1 - \sin x\cos x) &\equiv \sin x - \sin^2 x\cos x + \cos x - \sin x\cos^2 x \\ &\equiv \sin x - (1 - \cos^2 x)\cos x + \cos x - \sin x(1 - \sin^2 x) \\ &\equiv \sin x - \cos x + \cos^3 x + \cos x - \sin x + \sin^3 x \\ &\equiv \sin^3 x + \cos^3 x \quad \text{:LHS.}\end{aligned}$$

37 c)

LHS:

$$\begin{aligned}1 + \tan^2 x \equiv 1 + \frac{\sin^2 x}{\cos^2 x} &\equiv \frac{\cos^2 x}{\cos^2 x} + \frac{\sin^2 x}{\cos^2 x} \\ &\equiv \frac{\cos^2 x + \sin^2 x}{\cos^2 x} \equiv \frac{1}{\cos^2 x} \quad \text{:RHS.}\end{aligned}$$

37 d)

LHS:

$$\frac{1}{\tan^2 x} + 1 \equiv \frac{1}{\frac{\sin^2 x}{\cos^2 x}} + 1 \equiv \frac{\cos^2 x}{\sin^2 x} + 1$$
$$\equiv \frac{\cos^2 x}{\sin^2 x} + \frac{\sin^2 x}{\sin^2 x} \equiv \frac{\cos^2 x + \sin^2 x}{\sin^2 x} \equiv \frac{1}{\sin^2 x} \quad \text{:RHS.}$$

38 a)

LHS: $\frac{\cos x}{\sin x} + \frac{\sin x}{\cos x} \equiv \frac{\cos^2 x + \sin^2 x}{\sin x \cos x} \equiv \frac{1}{\sin x \cos x}$:RHS.

38 b)

LHS:

$$\frac{1}{\cos x + 1} - \frac{1}{\cos x - 1} \equiv \frac{(\cos x - 1) - (\cos x + 1)}{(\cos x + 1)(\cos x - 1)} \equiv \frac{-2}{\cos^2 x - 1}$$
$$\equiv \frac{2}{1 - \cos^2 x} \equiv \frac{2}{\sin^2 x} \quad \text{:RHS.}$$

38 c)

LHS:

$$\frac{\cos x}{1 - \sin x} + \frac{\cos x}{1 + \sin x} \equiv \frac{\cos x(1 + \sin x) + \cos x(1 - \sin x)}{(1 - \sin x)(1 + \sin x)}$$
$$\equiv \frac{\cos x + \sin x \cos x + \cos x - \sin x \cos x}{1 - \sin^2 x}$$
$$\equiv \frac{2\cos x}{1 - \sin^2 x} \equiv \frac{2\cos x}{\cos^2 x} \equiv \frac{2}{\cos x} \quad \text{:RHS.}$$

38 d)

LHS:

$$\frac{\cos x}{\sin x - 1} + \frac{\cos x}{\sin x + 1} \equiv \frac{\cos x(\sin x + 1) + \cos x(\sin x - 1)}{(\sin x - 1)(\sin x + 1)}$$
$$\equiv \frac{\sin x \cos x + \cos x + \sin x \cos x - \cos x}{\sin^2 x - 1} \equiv \frac{2\sin x \cos x}{\sin^2 x - 1}$$
$$\equiv -\frac{2\sin x \cos x}{1 - \sin^2 x} \equiv -\frac{2\sin x \cos x}{\cos^2 x} \equiv -\frac{2\sin x}{\cos x}$$
$$\equiv -2\tan x \quad \text{:RHS.}$$

39 a)

LHS:

$$\frac{1}{1 + \tan^2 x} \equiv \frac{1}{1 + \frac{\sin^2 x}{\cos^2 x}} \equiv \frac{\cos^2 x}{\cos^2 x + \sin^2 x} \equiv \frac{\cos^2 x}{1}$$
$$\equiv \cos^2 x \quad \text{:RHS.}$$

39 b)

LHS:

$$\frac{\tan x}{1+\tan^2 x} \equiv \frac{\frac{\sin x}{\cos x}}{1+\frac{\sin^2 x}{\cos^2 x}} \equiv \frac{\sin x \cos x}{\cos^2 x + \sin^2 x} \equiv \frac{\sin x \cos x}{1}$$
$$\equiv \sin x \cos x \quad \text{:RHS.}$$

39 c)

LHS:

$$\frac{1}{1-\tan^2 x} \equiv \frac{1}{1-\frac{\sin^2 x}{\cos^2 x}} \equiv \frac{\cos^2 x}{\cos^2 x - \sin^2 x}$$
$$\equiv \frac{\cos^2 x}{(1-\sin^2 x)-\sin^2 x} \equiv \frac{\cos^2 x}{1-2\sin^2 x} \quad \text{:RHS.}$$

39 d)

RHS:

$$\frac{1}{1-\tan^4 x} \equiv \frac{1}{(1-\tan^2 x)(1+\tan^2 x)}$$
$$\equiv \frac{1}{(1-\tan^2 x)(1+\frac{\sin^2 x}{\cos^2 x})} \equiv \frac{1}{(1-\tan^2 x)(\frac{\cos^2 x}{\cos^2 x}+\frac{\sin^2 x}{\cos^2 x})}$$
$$\equiv \frac{1}{(1-\tan^2 x)(\frac{\cos^2 x+\sin^2 x}{\cos^2 x})} \equiv \frac{1}{(1-\tan^2 x)\times\frac{1}{\cos^2 x}}$$
$$\equiv \frac{\cos^2 x}{1-\tan^2 x} \quad \text{:LHS.}$$

40 a)

LHS:

$$\frac{1}{\frac{1}{\sin^2 x}-1} \equiv \frac{1}{\frac{1}{\sin^2 x}-\frac{\sin^2 x}{\sin^2 x}} \equiv \frac{1}{\frac{1-\sin^2 x}{\sin^2 x}} \equiv \frac{1}{\frac{\cos^2 x}{\sin^2 x}} \equiv \frac{\sin^2 x}{\cos^2 x}$$
$$\equiv \tan^2 x \quad \text{:RHS.}$$

40 b)

LHS:

$$\frac{1}{\frac{1}{\cos^2 x}-1} \equiv \frac{1}{\frac{1}{\cos^2 x}-\frac{\cos^2 x}{\cos^2 x}} \equiv \frac{1}{\frac{1-\cos^2 x}{\cos^2 x}} \equiv \frac{1}{\frac{\sin^2 x}{\cos^2 x}} \equiv \frac{\cos^2 x}{\sin^2 x}$$
$$\equiv \frac{1-\sin^2 x}{\sin^2 x} \equiv \frac{1}{\sin^2 x}-\frac{\sin^2 x}{\sin^2 x} \equiv \frac{1}{\sin^2 x}-1 \quad \text{:RHS.}$$

40 c)

LHS:

$$\frac{1}{\frac{1}{\tan^2 x}+1} \equiv \frac{1}{\frac{1}{\frac{\sin^2 x}{\cos^2 x}}+1} \equiv \frac{1}{\frac{\cos^2 x}{\sin^2 x}+1} \equiv \frac{1}{\frac{\cos^2 x}{\sin^2 x}+\frac{\sin^2 x}{\sin^2 x}}$$
$$\equiv \frac{1}{\frac{\cos^2 x+\sin^2 x}{\sin^2 x}} \equiv \frac{1}{\frac{1}{\sin^2 x}} \equiv \sin^2 x \quad \text{:RHS.}$$

40 d)

LHS:

$$\frac{1}{\frac{1}{\sin^2 x}+\frac{1}{\cos^2 x}} \equiv \frac{1}{\frac{\cos^2 x+\sin^2 x}{\sin^2 x\cos^2 x}} \equiv \frac{1}{\frac{1}{\sin^2 x\cos^2 x}}$$
$$\equiv \sin^2 x\cos^2 x \equiv (\sin x\cos x)^2 \quad \text{:RHS.}$$

41 a)

LHS:

$$\sin^4 x-\cos^4 x \equiv (\sin^2 x+\cos^2 x)(\sin^2 x-\cos^2 x)$$
$$\equiv 1\times(\sin^2 x-\cos^2 x) \equiv \sin^2 x-\cos^2 x \quad \text{:RHS.}$$

41 b)

LHS:

$$\sin^2 x+\frac{\sin^4 x}{\cos^2 x} \equiv \frac{\sin^2 x\cos^2 x}{\cos^2 x}+\frac{\sin^4 x}{\cos^2 x}$$
$$\equiv \frac{\sin^2 x\cos^2 x+\sin^4 x}{\cos^2 x} \equiv \frac{\sin^2 x(\cos^2 x+\sin^2 x)}{\cos^2 x} \equiv \frac{\sin^2 x\times 1}{\cos^2 x}$$
$$\equiv \frac{\sin^2 x}{\cos^2 x} \equiv \tan^2 x \quad \text{:RHS.}$$

41 c)

LHS:

$$\cos^4 x+\sin^2 x+\sin^2 x\cos^2 x+\sin^2 x\tan^2 x$$
$$\equiv \cos^4 x+\sin^2 x+(1-\cos^2 x)\cos^2 x+\sin^2 x\times\frac{\sin^2 x}{\cos^2 x}$$
$$\equiv \cos^4 x+\sin^2 x+\cos^2 x-\cos^4 x+\frac{\sin^4 x}{\cos^2 x}$$
$$\equiv (\sin^2 x+\cos^2 x)+\frac{\sin^4 x}{\cos^2 x} \equiv 1+\frac{\sin^4 x}{\cos^2 x}$$
$$\equiv \frac{\cos^2 x}{\cos^2 x}+\frac{\sin^4 x}{\cos^2 x} \equiv \frac{\cos^2 x+\sin^4 x}{\cos^2 x} \equiv \frac{\cos^2 x+(1-\cos^2 x)^2}{\cos^2 x}$$
$$\equiv \frac{\cos^2 x+(1-2\cos^2 x+\cos^4 x)}{\cos^2 x} \equiv \frac{\cos^4 x-\cos^2 x+1}{\cos^2 x}$$
$$\equiv \frac{\cos^4 x}{\cos^2 x}+\frac{1-\cos^2 x}{\cos^2 x} \equiv \cos^2 x+\frac{\sin^2 x}{\cos^2 x}$$
$$\equiv \cos^2 x+\tan^2 x \quad \text{:RHS.}$$

41 d)

LHS:

$$\sin^2 x\cos^2 x-1+\sin^4 x-\tan^2 x \equiv \sin^2 x(1-\sin^2 x)-1+\sin^4 x-\frac{\sin^2 x}{\cos^2 x}$$
$$\equiv \sin^2 x-\sin^4 x-1+\sin^4 x-\frac{1-\cos^2 x}{\cos^2 x}$$

$$\equiv \sin^2 x - 1 - \left(\frac{1}{\cos^2 x} - \frac{\cos^2 x}{\cos^2 x}\right)$$
$$\equiv \sin^2 x - 1 - \frac{1}{\cos^2 x} + 1 \equiv \sin^2 x - \frac{1}{\cos^2 x} \quad \text{:RHS.}$$

42 a)

LHS:

$$\frac{\sin x}{1-\tan x} + \frac{\cos x}{1-\frac{1}{\tan x}} \equiv \frac{\sin x}{1-\frac{\sin x}{\cos x}} + \frac{\cos x}{1-\frac{1}{\frac{\sin x}{\cos x}}}$$
$$\equiv \frac{\sin x}{1-\frac{\sin x}{\cos x}} + \frac{\cos x}{1-\frac{\cos x}{\sin x}} \equiv \frac{\sin x \cos x}{\cos x - \sin x} + \frac{\sin x \cos x}{\sin x - \cos x}$$
$$\equiv \frac{\sin x \cos x}{\cos x - \sin x} - \frac{\sin x \cos x}{\cos x - \sin x} \equiv 0 \quad \text{:RHS.}$$

42 b)

LHS:

$$\frac{\cos x}{1-\tan x} + \frac{\sin x}{1-\frac{1}{\tan x}} \equiv \frac{\cos x}{1-\frac{\sin x}{\cos x}} + \frac{\sin x}{1-\frac{1}{\frac{\sin x}{\cos x}}}$$
$$\equiv \frac{\cos x}{1-\frac{\sin x}{\cos x}} + \frac{\sin x}{1-\frac{\cos x}{\sin x}} \equiv \frac{\cos^2 x}{\cos x - \sin x} + \frac{\sin^2 x}{\sin x - \cos x}$$
$$\equiv \frac{\cos^2 x}{\cos x - \sin x} - \frac{\sin^2 x}{\cos x - \sin x} \equiv \frac{\cos^2 x - \sin^2 x}{\cos x - \sin x}$$
$$\equiv \frac{(\cos x + \sin x)(\cos x - \sin x)}{\cos x - \sin x} \equiv \cos x + \sin x \quad \text{:RHS.}$$

42 c)

RHS:

$$\frac{2\sin x - 2\cos x}{2 - \frac{1}{\sin x \cos x}} \equiv \frac{2\sin^2 x \cos x - 2\sin x \cos^2 x}{2\sin x \cos x - 1}$$
$$\equiv \frac{2\sin^2 x \cos x - 2\sin x \cos^2 x}{2\sin x \cos x - (\sin^2 x + \cos^2 x)}$$
$$\equiv \frac{2\sin^2 x \cos x - 2\sin x \cos^2 x}{2\sin x \cos x - \sin^2 x - \cos^2 x}$$
$$\equiv \frac{2\sin x \cos x(\sin x - \cos x)}{(\cos x - \sin x)(\sin x - \cos x)} \equiv \frac{2\sin x \cos x}{\cos x - \sin x}$$
$$\equiv \frac{\sin x \cos x}{\cos x - \sin x} + \frac{\sin x \cos x}{\cos x - \sin x} \equiv \frac{\sin x \cos x}{\cos x - \sin x} - \frac{\sin x \cos x}{\sin x - \cos x}$$
$$\equiv \frac{\sin x}{1-\frac{\sin x}{\cos x}} - \frac{\cos x}{1-\frac{\cos x}{\sin x}} \equiv \frac{\sin x}{1-\frac{\sin x}{\cos x}} - \frac{\cos x}{1-\frac{1}{\frac{\sin x}{\cos x}}}$$
$$\equiv \frac{\sin x}{1-\tan x} - \frac{\cos x}{1-\frac{1}{\tan x}} \quad \text{:LHS.}$$

42 d)

LHS:

$$\frac{\cos x}{1-\tan x}-\frac{\sin x}{1-\frac{1}{\tan x}} \equiv \frac{\cos x}{1-\frac{\sin x}{\cos x}}-\frac{\sin x}{1-\frac{1}{\frac{\sin x}{\cos x}}}$$
$$\equiv \frac{\cos x}{1-\frac{\sin x}{\cos x}}-\frac{\sin x}{1-\frac{\cos x}{\sin x}} \equiv \frac{\cos^2 x}{\cos x-\sin x}-\frac{\sin^2 x}{\sin x-\cos x}$$
$$\equiv \frac{\cos^2 x}{\cos x-\sin x}+\frac{\sin^2 x}{\cos x-\sin x} \equiv \frac{\cos^2 x+\sin^2 x}{\cos x-\sin x} \equiv \frac{1}{\cos x-\sin x} \quad \text{:RHS.}$$

43 a)

LHS:

$$\frac{\sin x}{1-\cos x} \equiv \frac{\sin x(1+\cos x)}{(1-\cos x)(1+\cos x)} \equiv \frac{\sin x(1+\cos x)}{1-\cos^2 x}$$
$$\equiv \frac{\sin x(1+\cos x)}{\sin^2 x} \equiv \frac{1+\cos x}{\sin x} \quad \text{:RHS.}$$

43 b)

LHS:

$$\frac{\cos x}{1-\sin x} \equiv \frac{\cos x(1+\sin x)}{(1-\sin x)(1+\sin x)} \equiv \frac{\cos x(1+\sin x)}{1-\sin^2 x}$$
$$\equiv \frac{\cos x(1+\sin x)}{\cos^2 x} \equiv \frac{1+\sin x}{\cos^2 x} \quad \text{:RHS.}$$

43 c)

LHS: $\frac{\sin x}{1-\tan x} \equiv \frac{\sin x}{1-\frac{\sin x}{\cos x}} \equiv \frac{\sin x \cos x}{\cos x-\sin x}$:RHS.

43 d)

RHS:

$$\frac{1+\cos x}{\sin^2 x}-\frac{1+\cos x}{\sin x} \equiv \frac{1+\cos x}{\sin^2 x}-\frac{(1+\cos x)\sin x}{\sin^2 x}$$
$$\equiv \frac{(1+\cos x)-(1+\cos x)\sin x}{\sin^2 x} \equiv \frac{(1+\cos x)(1-\sin x)}{1-\cos^2 x}$$
$$\equiv \frac{(1+\cos x)(1-\sin x)}{(1-\cos x)(1+\cos x)} \equiv \frac{1-\sin x}{1-\cos x} \quad \text{:LHS.}$$

44 a)

LHS:

$$\frac{\tan x+\frac{1}{\tan x}}{\frac{1}{\cos x}+\frac{1}{\sin x}} \equiv \frac{\frac{\sin x}{\cos x}+\frac{1}{\frac{\sin x}{\cos x}}}{\frac{1}{\cos x}+\frac{1}{\sin x}} \equiv \frac{\frac{\sin x}{\cos x}+\frac{\cos x}{\sin x}}{\frac{1}{\cos x}+\frac{1}{\sin x}} \equiv \frac{\sin x+\frac{\cos^2 x}{\sin x}}{1+\frac{\cos x}{\sin x}}$$
$$\equiv \frac{\sin^2 x+\cos^2 x}{\sin x+\cos x} \equiv \frac{1}{\sin x+\cos x} \quad \text{:RHS.}$$

44 b)

LHS:

$$\frac{\cos x - \frac{1}{\cos x}}{\frac{1}{\sin x} - \frac{1}{\cos x}} \equiv \frac{\cos^2 x - 1}{\frac{\cos x}{\sin x} - 1} \equiv \frac{1 - \cos^2 x}{1 - \frac{\cos x}{\sin x}} \equiv \frac{\sin^2 x}{1 - \frac{1}{\frac{\sin x}{\cos x}}}$$

$$\equiv \frac{\sin^2 x}{1 - \frac{1}{\tan x}} \quad \text{:RHS.}$$

44 c)

LHS:

$$\frac{\sin x - \frac{1}{\sin x}}{\frac{1}{\sin x} - \frac{1}{\cos x}} \equiv \frac{\sin^2 x - 1}{1 - \frac{\sin x}{\cos x}} \equiv \frac{1 - \sin^2 x}{\frac{\sin x}{\cos x} - 1} \equiv \frac{\cos^2 x}{\frac{\sin x}{\cos x} - 1}$$

$$\equiv \frac{\cos^3 x}{\sin x - \cos x} \quad \text{:RHS.}$$

45 a)

RHS:

$$\frac{1}{\frac{1}{\sin x} + \frac{1}{\cos x} + \tan x} \equiv \frac{1}{\frac{1}{\sin x} + \frac{1}{\cos x} + \frac{\sin x}{\cos x}}$$

$$\equiv \frac{1}{\frac{\cos x}{\sin x \cos x} + \frac{\sin x}{\sin x \cos x} + \frac{\sin^2 x}{\sin x \cos x}} \equiv \frac{1}{\frac{\cos x + \sin x + \sin^2 x}{\sin x \cos x}}$$

$$\equiv \frac{\sin x \cos x}{\cos x + \sin x + \sin^2 x} \equiv \frac{\sin x \cos x}{\cos x + \sin x + (1 - \cos^2 x)}$$

$$\equiv \frac{\sin x \cos x}{1 + \sin x + \cos x - \cos^2 x} \quad \text{:LHS.}$$

45 b)

RHS:

$$\frac{1}{\frac{\tan x}{\cos x} - \frac{1}{\sin x} - \tan x} \equiv \frac{1}{\frac{\frac{\sin x}{\cos x}}{\cos x} - \frac{1}{\sin x} - \frac{\sin x}{\cos x}} \equiv \frac{1}{\frac{\sin x}{\cos^2 x} - \frac{1}{\sin x} - \frac{\sin x}{\cos x}}$$

$$\equiv \frac{1}{\frac{\sin^2 x}{\sin x \cos^2 x} - \frac{\cos^2 x}{\sin x \cos^2 x} - \frac{\sin^2 x \cos x}{\sin x \cos^2 x}}$$

$$\equiv \frac{1}{\frac{\sin^2 x - \cos^2 x - \sin^2 x \cos x}{\sin x \cos^2 x}} \equiv \frac{\sin x \cos^2 x}{\sin^2 x - \cos^2 x - \sin^2 x \cos x}$$

$$\equiv \frac{\sin x \cos^2 x}{(1 - \cos^2 x) - \cos^2 x - (1 - \cos^2 x)\cos x}$$

$$\equiv \frac{\sin x \cos^2 x}{1 - \cos^2 x - \cos^2 x - \cos x + \cos^3 x} \equiv \frac{\sin x \cos^2 x}{\cos^3 x - 2\cos^2 x - \cos x + 1} \quad \text{:LHS.}$$

46 a)

LHS:

$$\left(\frac{1}{\cos x} - \sin x\right)\left(\frac{1}{\sin x} - \cos x\right)$$

$$\equiv \left(\frac{1}{\cos x} - \frac{\sin x \cos x}{\cos x}\right)\left(\frac{1}{\sin x} - \frac{\sin x \cos x}{\sin x}\right)$$
$$\equiv \left(\frac{1-\sin x \cos x}{\cos x}\right)\left(\frac{1-\sin x \cos x}{\sin x}\right)$$
$$\equiv \frac{(1-\sin x \cos x)^2}{\sin x \cos x} \equiv \frac{(1-\sin x \cos x)^2}{\sin x \cos x} \equiv \frac{(\sin x \cos x - 1)^2}{\sin x \cos x} \quad \text{:RHS.}$$

46 b)

LHS:

$$\left(\frac{1}{\sin x} - \sin x\right)\left(\frac{1}{\cos x} - \cos x\right)$$
$$\equiv \left(\frac{1}{\sin x} - \frac{\sin^2 x}{\sin x}\right)\left(\frac{1}{\cos x} - \frac{\cos^2 x}{\cos x}\right)$$
$$\equiv \left(\frac{1-\sin^2 x}{\sin x}\right)\left(\frac{1-\cos^2 x}{\cos x}\right) \equiv \frac{\cos^2 x}{\sin x}\left(\frac{1-\cos^2 x}{\cos x}\right)$$
$$\equiv \frac{\cos^2 x - \cos^4 x}{\sin x \cos x} \equiv \frac{1-\cos^2 x}{\frac{\sin x}{\cos x}} \equiv \frac{\sin^2 x}{\frac{\sin x}{\cos x}} \equiv \frac{\sin^2 x}{1} \times \frac{\cos x}{\sin x}$$
$$\equiv \sin x \cos x \equiv \frac{1}{\frac{1}{\sin x \cos x}} \equiv \frac{1}{\frac{\sin^2 x + \cos^2 x}{\sin x \cos x}}$$
$$\equiv \frac{1}{\frac{\sin^2 x}{\sin x \cos x} + \frac{\cos^2 x}{\sin x \cos x}} \equiv \frac{1}{\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}} \equiv \frac{1}{\frac{\sin x}{\cos x} + \frac{1}{\frac{\sin x}{\cos x}}}$$
$$\equiv \frac{1}{\tan x + \frac{1}{\tan x}} \quad \text{:RHS.}$$

47)

RHS:

$$\frac{1}{\sin x - \cos x + \frac{2\cos^2 x}{\sin x + \cos x}} \equiv \frac{\sin x + \cos x}{(\sin x - \cos x)(\sin x + \cos x) + 2\cos^2 x}$$
$$\equiv \frac{\sin x + \cos x}{\sin^2 x - \cos^2 x + 2\cos^2 x} \equiv \frac{\sin x + \cos x}{\sin^2 x + \cos^2 x} \equiv \sin x + \cos x$$
$$\equiv \frac{(\sin x + \cos x)(\cos x - \sin x)}{\cos x - \sin x}$$
$$\equiv \frac{\sin x \cos x - \sin^2 x + \cos^2 x - \sin x \cos x}{\cos x - \sin x} \equiv \frac{\cos^2 x - \sin^2 x}{\cos x - \sin x}$$
$$\equiv \frac{\cos^2 x - (1-\cos^2 x)}{\cos x - \sin x} \equiv \frac{2\cos^2 x - 1}{\cos x - \sin x} \quad \text{:LHS.}$$

48)

RHS:

$$\sin^2 A \sin^2 B - \cos^2 A \cos^2 B \equiv \sin^2 A \sin^2 B - (1-\sin^2 A)(1-\sin^2 B)$$
$$\equiv \sin^2 A \sin^2 B - (1 - \sin^2 A - \sin^2 B + \sin^2 A \sin^2 B)$$
$$\equiv \sin^2 A \sin^2 B - 1 + \sin^2 A + \sin^2 B - \sin^2 A \sin^2 B$$
$$\equiv \sin^2 A + \sin^2 B - 1 \quad \text{:RHS.}$$

49)

The first error is replacing $\cos^2 x$ with $\sin^2 x - 1$. She should instead have replaced $\cos^2 x$ with $1 - \sin^2 x$.

Note that Isla *can* divide through by $\sin x$ when she goes from $\sqrt{2}\sin^2 x + \sin x = 0$ to $\sqrt{2}\sin x + 1 = 0$ without losing solutions as $\sin x$ does not equal zero between $180°$ and $360°$ (exclusive), so this is not an error.

Isla's second mistake is in not finding solutions between $180°$ and $360°$, as she only finds $-45°$ which is not in the range we are looking for.

50 a)

$$\begin{aligned} 20\sin x \tan x + 31\sin x &= 7\cos x \\ \Rightarrow \frac{20\sin x \tan x}{\cos x} + \frac{31\sin x}{\cos x} &= \frac{7\cos x}{\cos x} \\ \Rightarrow 20\tan^2 x + 31\tan x &= 7 \\ \Rightarrow 20\tan^2 x + 31\tan x - 7 &= 0 \\ \Rightarrow (4\tan x + 7)(5\tan x - 1) &= 0. \end{aligned}$$

50 b)

$20\sin 4\theta \tan 4\theta + 31\sin 4\theta = 7\cos 4\theta$ for $0° \leq \theta < 45°$.

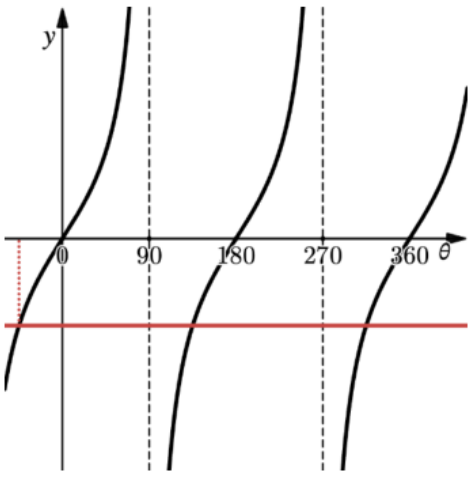
$\Rightarrow (4\tan 4\theta + 7)(5\tan 4\theta - 1) = 0.$

First solve $4\tan 4\theta + 7 = 0$ for $0° \leq \theta < 45°$.

$\Rightarrow \tan 4\theta = -\frac{7}{4}$.

Find first value:

$4\theta = \arctan(-\frac{7}{4}) = -60.2551\ldots°$ (not in range).



Apply transformation:

$4\theta = -60.2551\ldots° \Rightarrow \theta = -15.0637\ldots°$ (not in range).

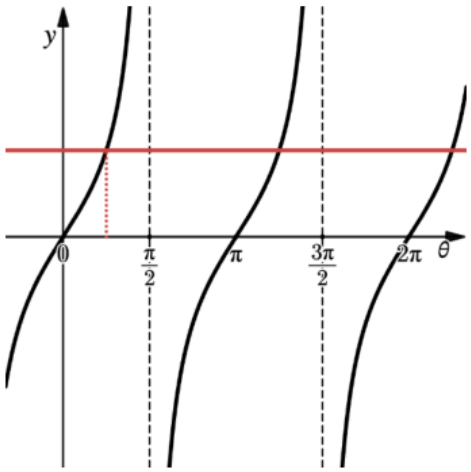
Add on period:

$\theta = -15.0637\ldots° + 45° = 29.9362\ldots°$ (in range).

Now solve $5\tan 4\theta - 1 = 0 \Rightarrow \tan 4\theta = \frac{1}{5}$.

Find first value:

$4\theta = \arctan\frac{1}{5} = 11.3099\ldots^\circ$ (in range).



Apply transformation: $4\theta = 11.3099\ldots^\circ \Rightarrow \theta = 2.8274\ldots$ (in range).

$\Rightarrow \theta = \{2.83^\circ, 29.9^\circ\}$ (3 SF).

51 a)

LHS:

$$\frac{\tan x}{\frac{1}{\cos x} - \cos x} \equiv \frac{\frac{\sin x}{\cos x}}{\frac{1}{\cos x} - \cos x} \equiv \frac{\sin x}{1 - \cos^2 x} \equiv \frac{\sin x}{\sin^2 x}$$
$$\equiv \frac{1}{\sin x} \quad \text{:RHS.}$$

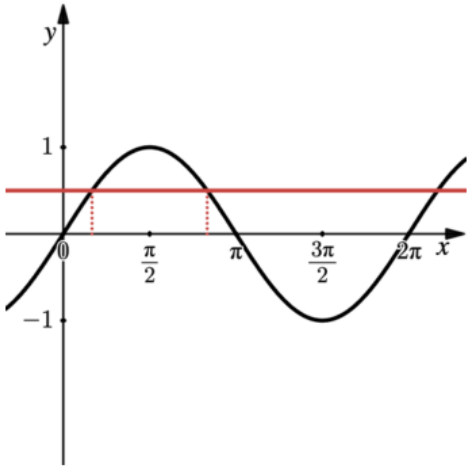
51 b)

$\dfrac{\tan x}{\frac{1}{\cos x} - \cos x} = 6$ for $0° \leq x < 360°$.

$\Rightarrow \dfrac{1}{\sin x} = 6 \Rightarrow \sin x = \frac{1}{6}$.

Find first value between $0°$ and $360°$:

$x = \arcsin \frac{1}{6} = 9.5940\ldots°$ (in range).



Find second value between $0°$ and $360°$:

$x = 180° - 9.5940\ldots° = 170.4059\ldots°$ (in range).

$x = \{9.59°, 170°\}$ (3 SF).

51 c)

$\dfrac{\tan(2\theta - 55°)}{\frac{1}{\cos(2\theta - 55°)} - \cos(2\theta - 55°)} = 6$ for $0° \leq \theta < 180°$.

$\Rightarrow \dfrac{1}{\sin(2\theta - 55°)} = 6 \Rightarrow \sin(2\theta - 55°) = \frac{1}{6}$.

Apply transformation:

$2\theta - 55° = 9.5940\ldots° \Rightarrow \theta = 32.2970\ldots$ (in range).

$2\theta - 55° = 170.4059\ldots° \Rightarrow \theta = 112.7029\ldots$ (in range).

$\Rightarrow \theta = \{32.3°, 113°\}$ (3 SF).

52 a)

LHS:

$$\frac{\cos^3 x}{\sin x} + \sin x \cos x \equiv \frac{\cos^3 x}{\sin x} + \frac{\sin^2 x \cos x}{\sin x}$$
$$\equiv \frac{\cos^3 x + \sin^2 x \cos x}{\sin x} \equiv \frac{\cos^2 x + \sin^2 x}{\frac{\sin x}{\cos x}} \equiv \frac{1}{\tan x} \quad \text{:RHS.}$$

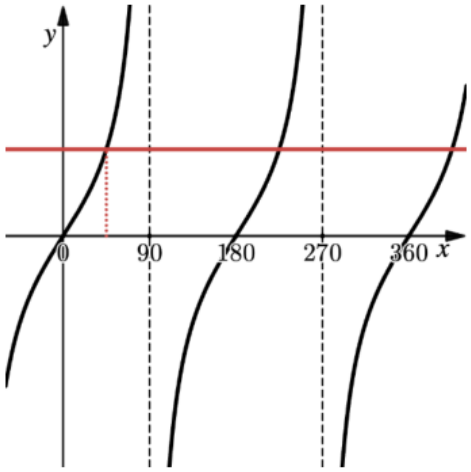
52 b)

$\frac{\cos^3 x}{\sin x} + \sin x \cos x = \frac{1}{20}$ for $0° \leq x < 360°$.

$\Rightarrow \frac{1}{\tan x} = \frac{1}{20} \Rightarrow \tan x = 20.$

Find first value:

$x = \arctan 20 = 87.137\ldots°.$



Add period:

$x = 87.137\ldots° + 180° = 267.137\ldots°$ (in range).

$x = \{87.1°, 267°\}$ (3 SF).

52 c)

$2\theta + 9° = 87.137\ldots° \Rightarrow \theta = 39.068\ldots°.$

$2\theta + 9° = 267.137\ldots° \Rightarrow \theta = 129.068\ldots°.$

$\Rightarrow \theta = \{39.1°, 129°\}$ (3 SF).

53)

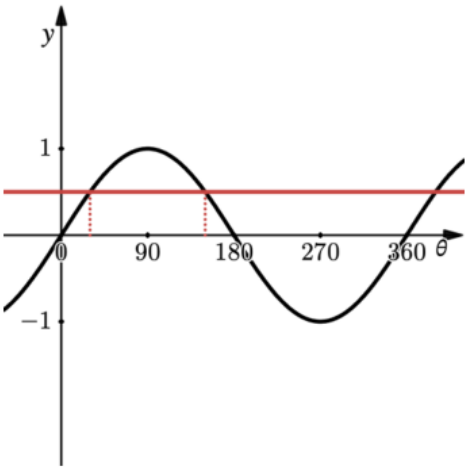
$\sin^4\theta - \cos^4\theta = \frac{1}{2}$ for $0° \le \theta \le 360°$.

$$\Rightarrow (\sin^2\theta + \cos^2\theta)(\sin^2\theta - \cos^2\theta) = \frac{1}{2}$$
$$\Rightarrow \sin^2\theta - \cos^2\theta = \frac{1}{2} \Rightarrow \sin^2\theta - (1 - \sin^2\theta) = \frac{1}{2}$$
$$\Rightarrow \sin^2\theta - 1 + \sin^2\theta = \frac{1}{2} \Rightarrow 2\sin^2\theta = \frac{3}{2}$$
$$\Rightarrow \sin^2\theta = \frac{3}{4} \Rightarrow \sin\theta = \pm\frac{\sqrt{3}}{2}.$$

First solve $\sin\theta = \frac{\sqrt{3}}{2}$ for $0° \le \theta \le 360°$.

Find first value between $0°$ and $360°$:

$\theta = \arcsin\frac{\sqrt{3}}{2} = 60°$ (in range).



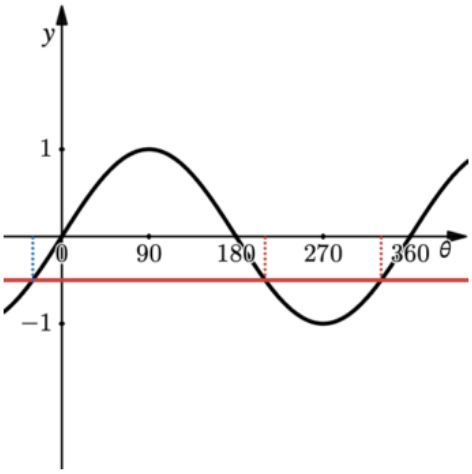
Find second value between $0°$ and $360°$:

$\theta = 180° - 60° = 120°$ (in range).

Now solve $\sin\theta = -\frac{\sqrt{3}}{2}$ for $0° \le \theta \le 360°$.

Find first value:

$\theta = \arcsin(-\frac{\sqrt{3}}{2}) = -60\ldots°$ (in range).



Find first value between $0°$ and $360°$:

$\theta = 180° + 60° = 240°$ (in range).

Find second value between $0°$ and $360°$:

$\theta = 360° - 60° = 300°$ (in range).

$\theta = \{60°, 120°, 240°, 300°\}$.

54)

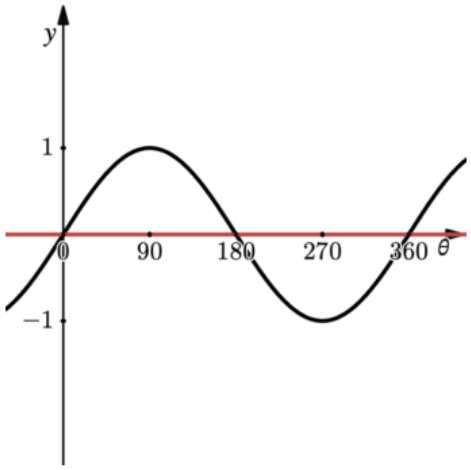
$2\cos^4\theta + \cos^2\theta\sin\theta - 2\cos^2\theta = 0$ for $0° \leq \theta \leq 360°$.

$$\begin{aligned}
&\Rightarrow 2(1-\sin^2\theta)^2 + (1-\sin^2\theta)\sin\theta - 2(1-\sin^2\theta) = 0\\
&\Rightarrow 2(1-2\sin^2\theta+\sin^4\theta) + \sin\theta - \sin^3\theta - 2 + 2\sin^2\theta = 0\\
&\Rightarrow 2 - 4\sin^2\theta + 2\sin^4\theta + \sin\theta - \sin^3\theta - 2 + 2\sin^2\theta = 0\\
&\Rightarrow 2\sin^4\theta - \sin^3\theta - 2\sin^2\theta + \sin\theta = 0\\
&\Rightarrow \sin\theta(2\sin\theta - 1)(\sin\theta - 1)(\sin\theta + 1) = 0.
\end{aligned}$$

First solve $\sin\theta = 0$ for $0° \leq \theta \leq 360°$.

Find first value:

$\theta = \arcsin 0 = 0°$ (in range).



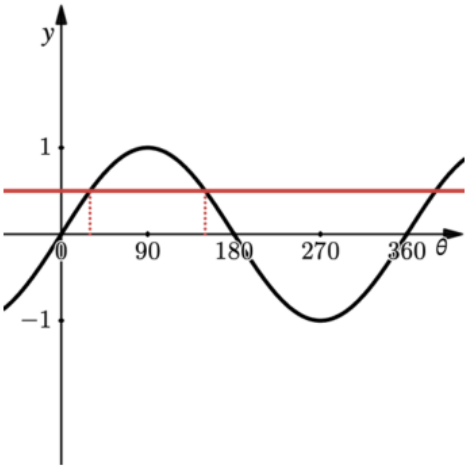
So $\theta = 180°$ (in range) and $\theta = 360°$ (in range).

Now solve $2\sin\theta - 1 = 0$ for $0° \leq \theta \leq 360°$.

$\Rightarrow \sin\theta = \frac{1}{2}$.

Find first value between $0°$ and $360°$:

$\theta = \arcsin\frac{1}{2} = 30°$ (in range).



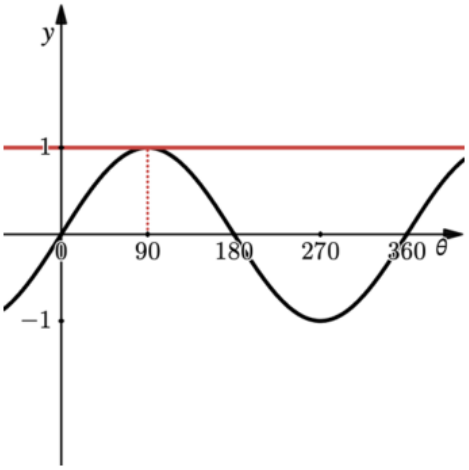
Find second value between $0°$ and $360°$:

$\theta = 180° - 30° = 150°$ (in range).

Now solve $\sin\theta - 1 = 0$ for $0° \leq \theta \leq 360°$.

$\Rightarrow \sin\theta = 1$.

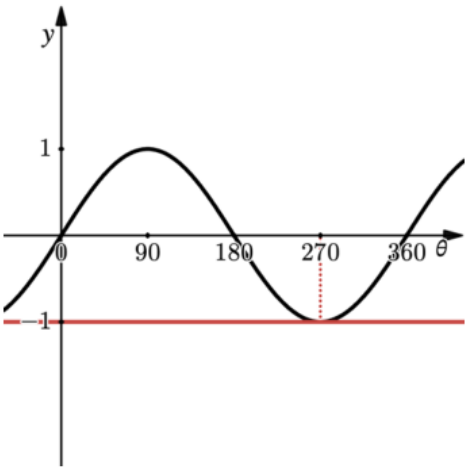
$\theta = \arcsin 1 = 90°$ (in range).



Now solve $\sin\theta + 1 = 0$ for $0° \leq \theta \leq 360°$.

$\Rightarrow \sin\theta = -1$.

$\theta = \arcsin(-1) = 270°$ (in range).



$\Rightarrow \theta = \{0°, 30°, 90°, 150°, 180°, 270°, 360°\}$.